

# Spatio-Temporal Strategic Modelling of Conflict Evolution

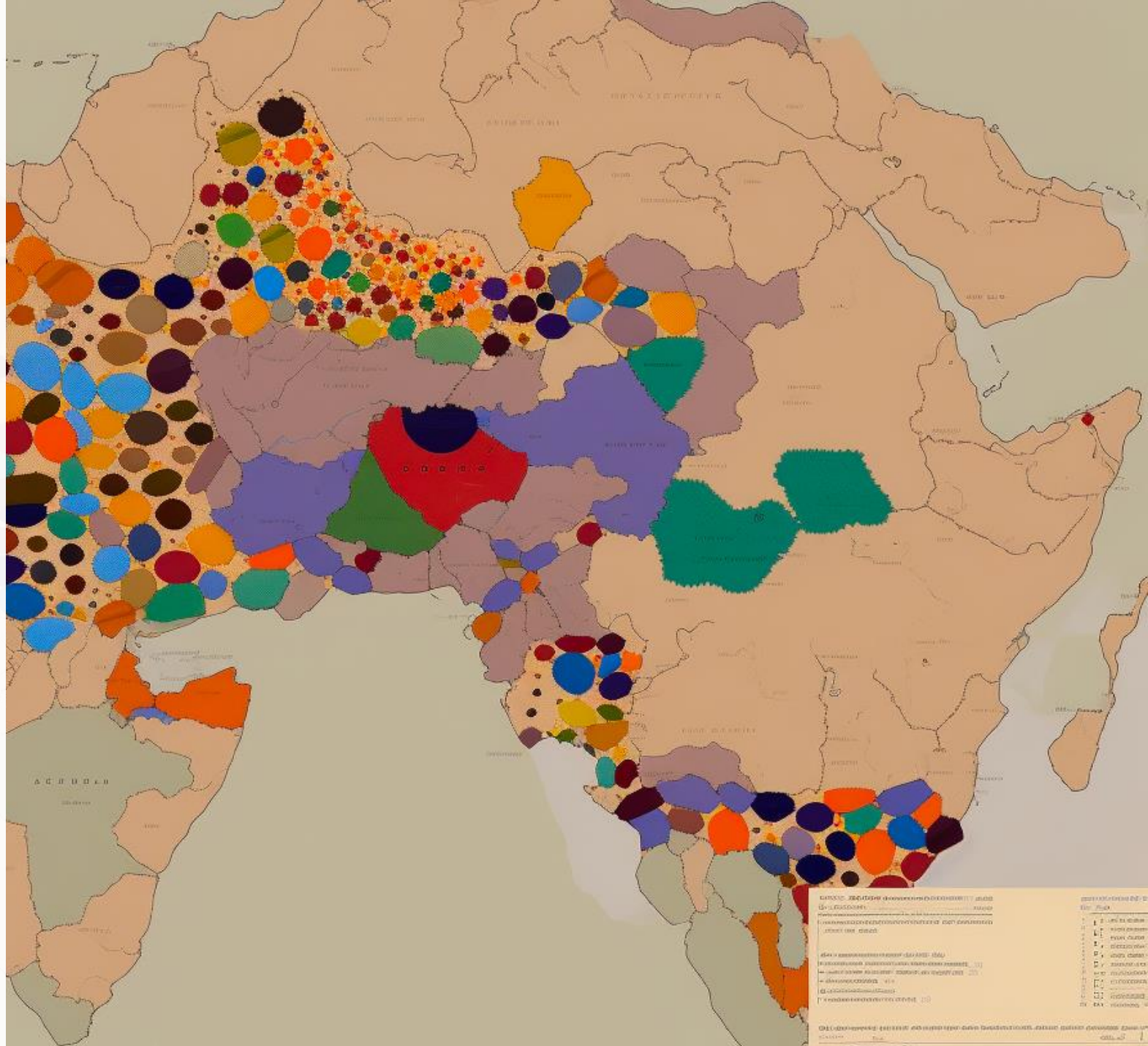
Vikram Iyer, Francesco Re, Lukas Rüttgers, Vincent Ventura





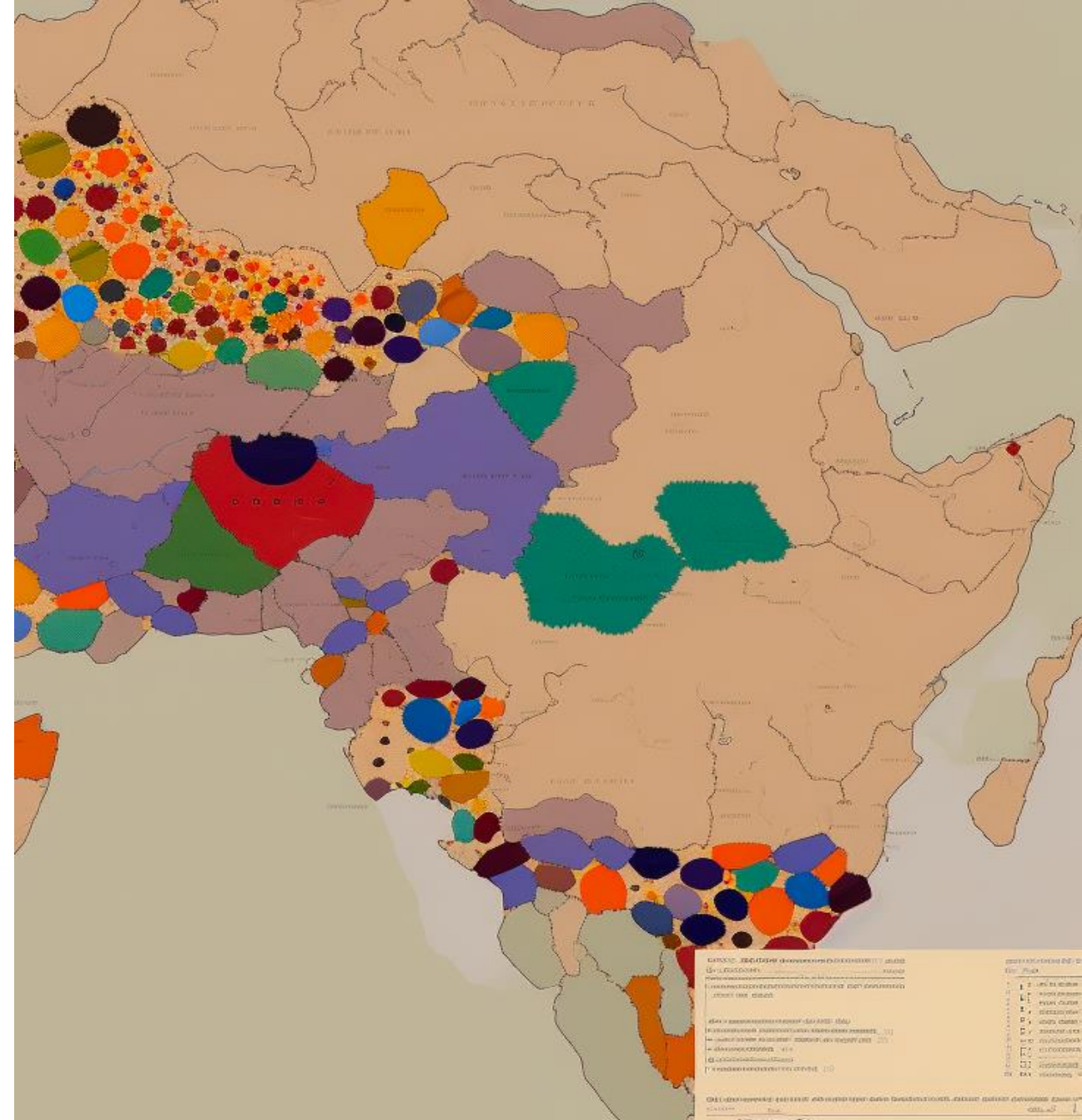
# Summary

1. Problem Statement
2. Motivation
3. Method
4. Analysis & Result



# 1. Problem Statement

- Armed conflict remains a persistent feature of the international landscape
  - It behaves like a complex system (Richardson 1948, 1960), (Cederman, 1997; 2003)
- Macro-level conflict patterns (escalation, diffusion, clustering) arise from micro-level decisions





# 1.1 Agent Based Modelling in Computational Conflict Research

## Epstein (Micro–Conflict Dynamics) [12]

Conflict emerges from **grievance vs opportunity**

Agents act based on:

- Local repression
- Perceived injustice
- Protest thresholds

Violence activates when **grievance exceeds control**

Emphasis:

- Individual behavior rules
- Local interaction structure
- Nonlinear escalation

## Cederman (Geo-Political Structure) [13]

Conflict shaped by **ethnic geography & state reach**

Groups embedded in:

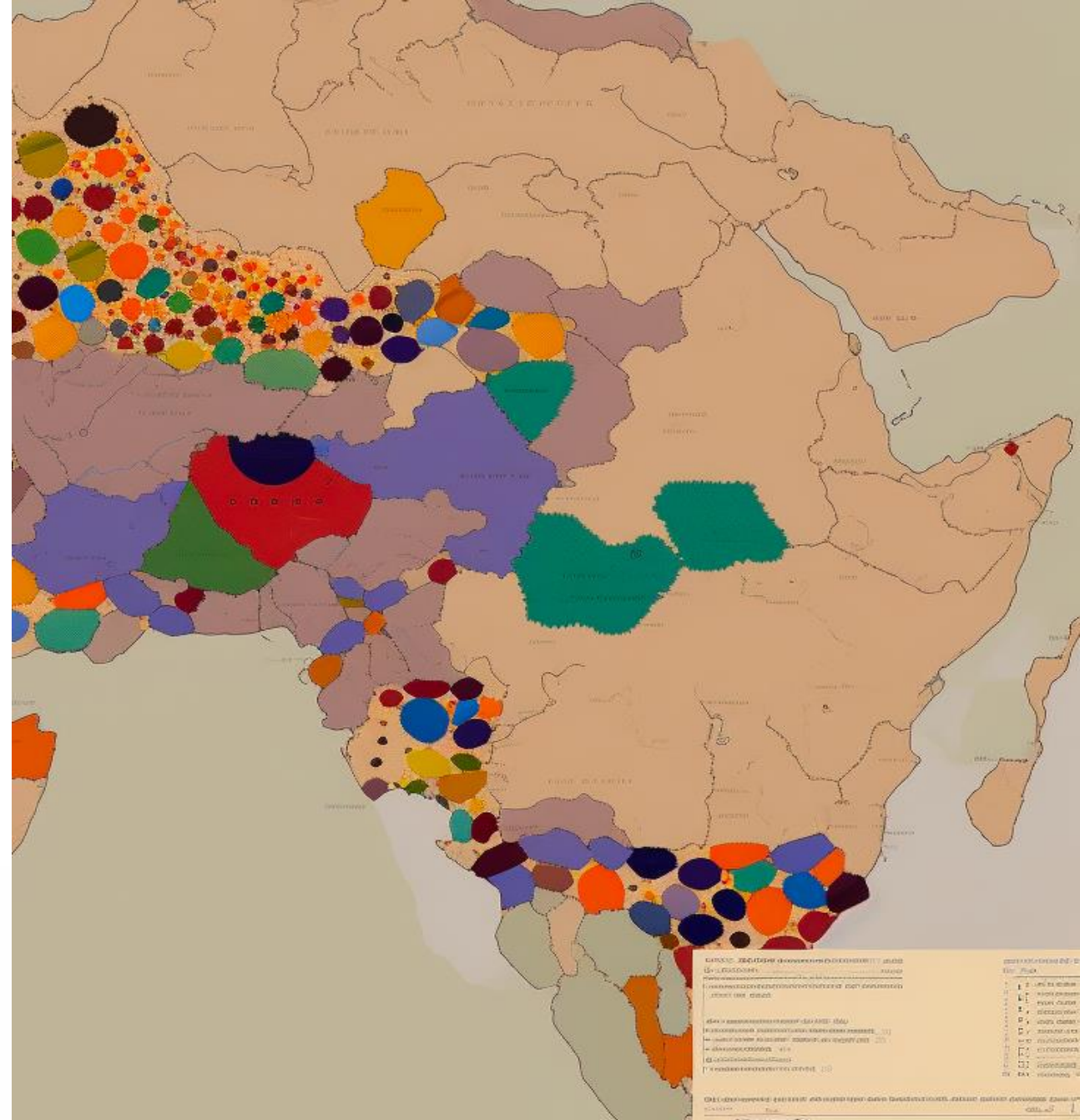
- Terrain
- Borders
- Population distributions

Emphasis:

- Spatial constraints
- Territorial fragmentation
- Center–periphery dynamics

## 2. Motivation

- We develop our method in light of the following dynamics in conflict:
  1. Armed groups establish areas of influence through territorial control.
  2. Interacting with each other and redefining their territories in accordance
  3. Their evolution can be understood as a spatio-temporal diffusion process



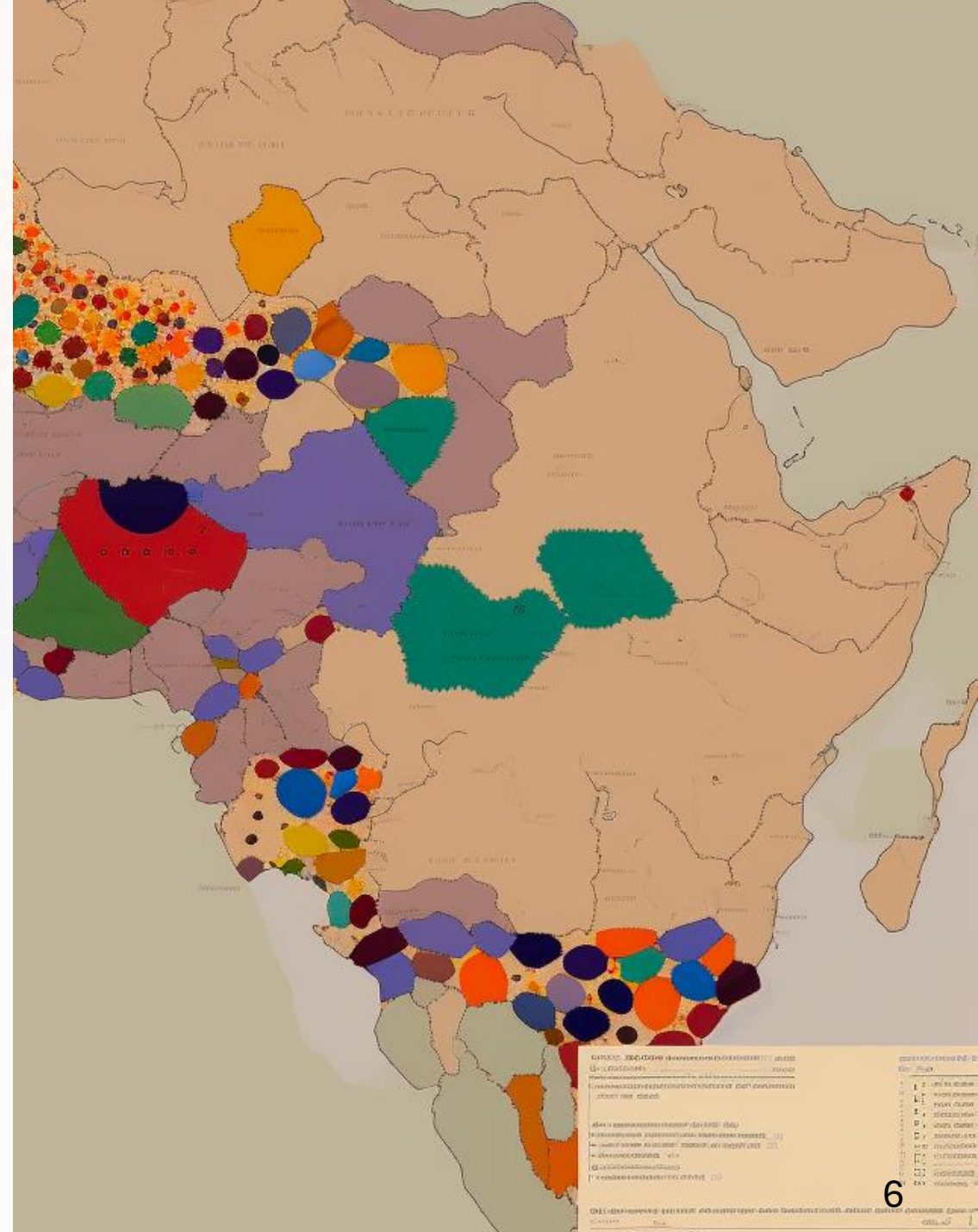


## 2.1 Armed groups establish areas of influence through territorial control.

- They are not individual entities but groups of people
  - The control they have over the territory is not binary, but differs from area to area
- We define armed groupss according to the area of influence they have over a territory

### *Desired Properties*

- Influence weakens with distance and strengthens around **resources, infrastructure, and population**
- Groups seek to **maximize territorial influence** by expanding into resource-rich regions.

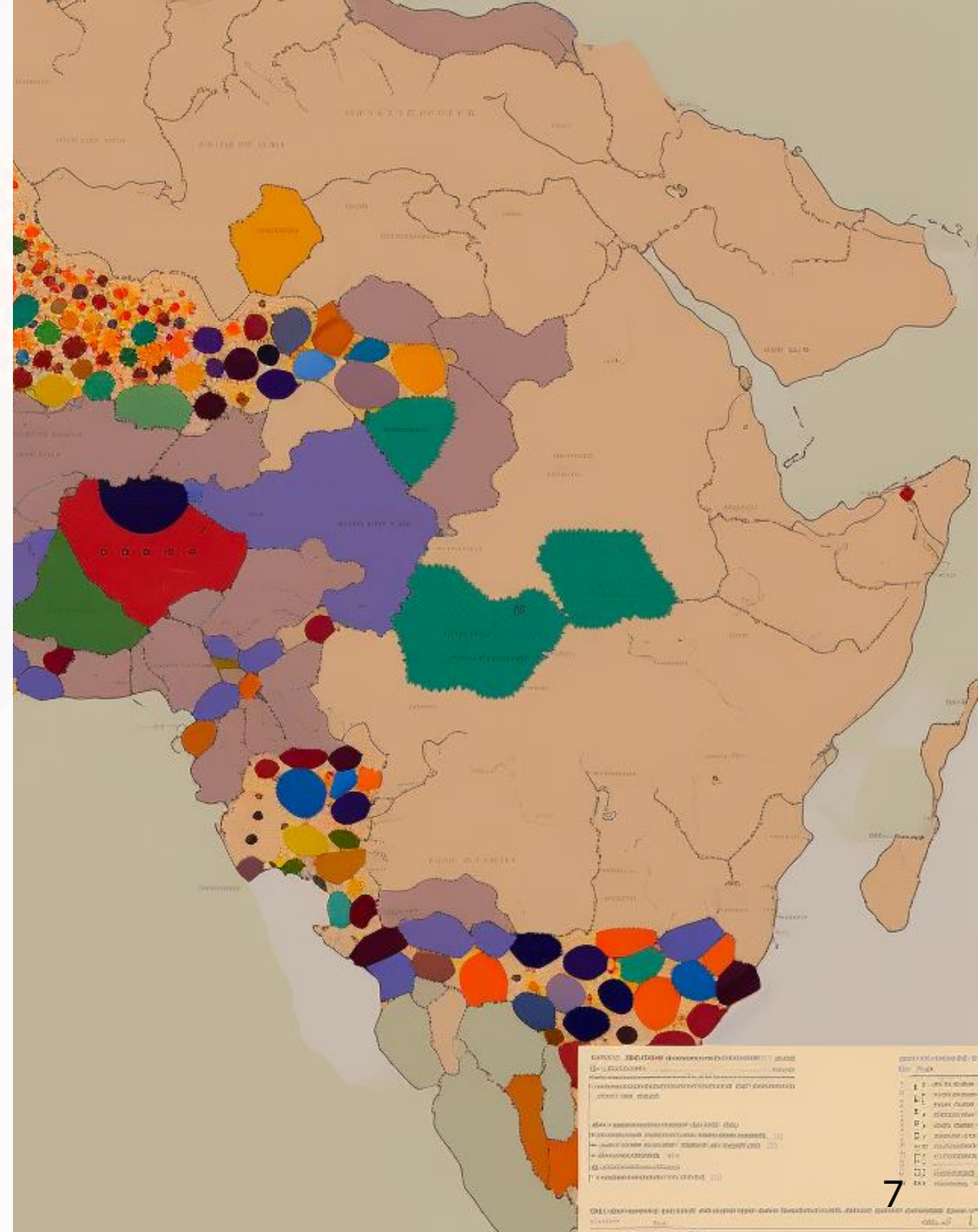


## 2.2 Armed groups interact with each other and define their territories in accordance.

- When expanding fields of influence collide, they create "strategic frontiers."

### *Desired Properties*

- Frontiers as interaction boundaries that allow the group to choose between **Attack** and **Cooperation**
- There are different possible strategies of interaction, e.g. 'Always-attack', 'Always-Cooperate', 'Tit-for-Tat'.



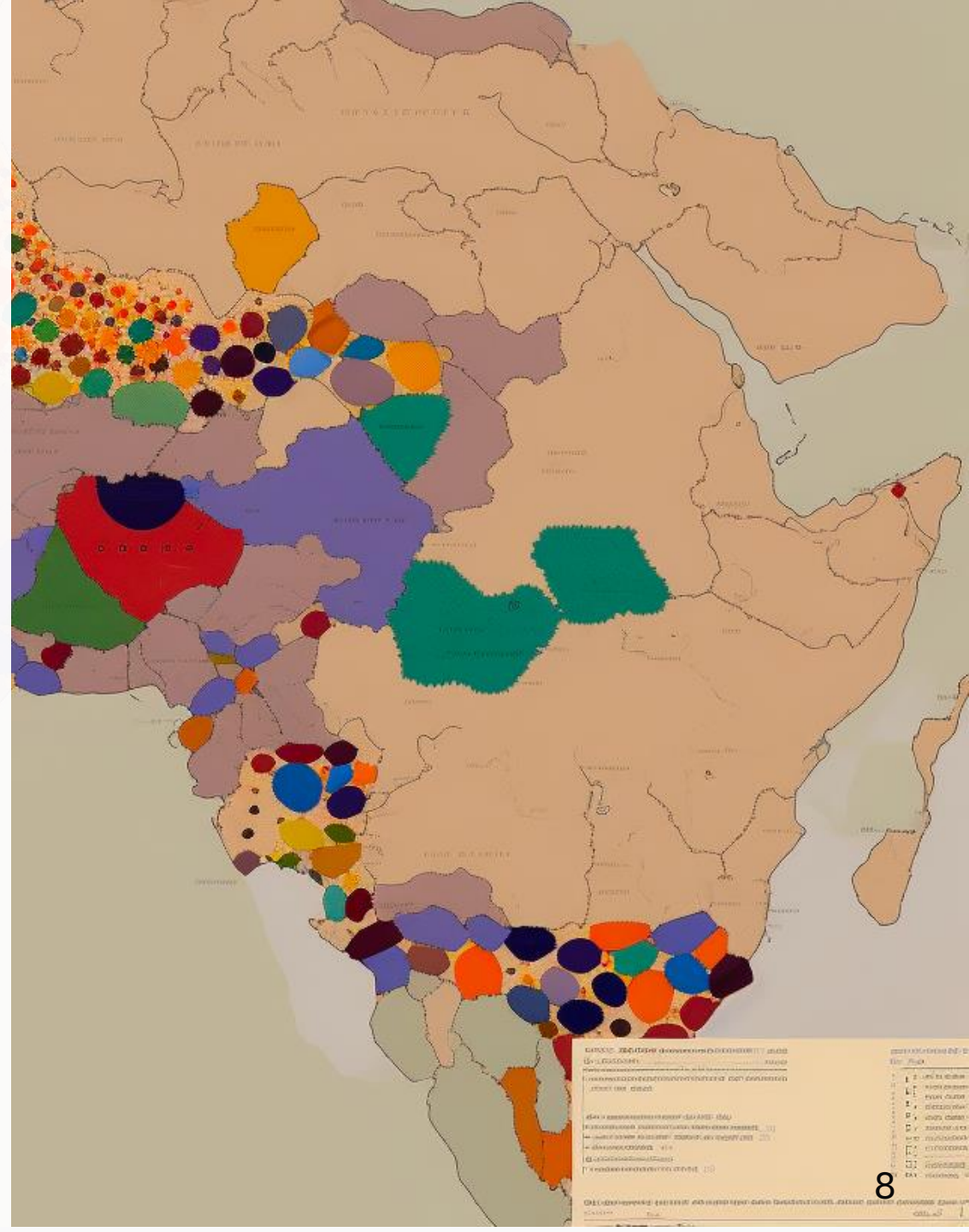


## 2.3 Their evolution can be understood as a spatio-temporal diffusion process

- Several conflict research scholars treat conflict as a physical system (e.g., Richardson, 1960)
- Evolution much akin to contagion processes studied in epidemiology (Beer, 1979)

### *Desired Properties*

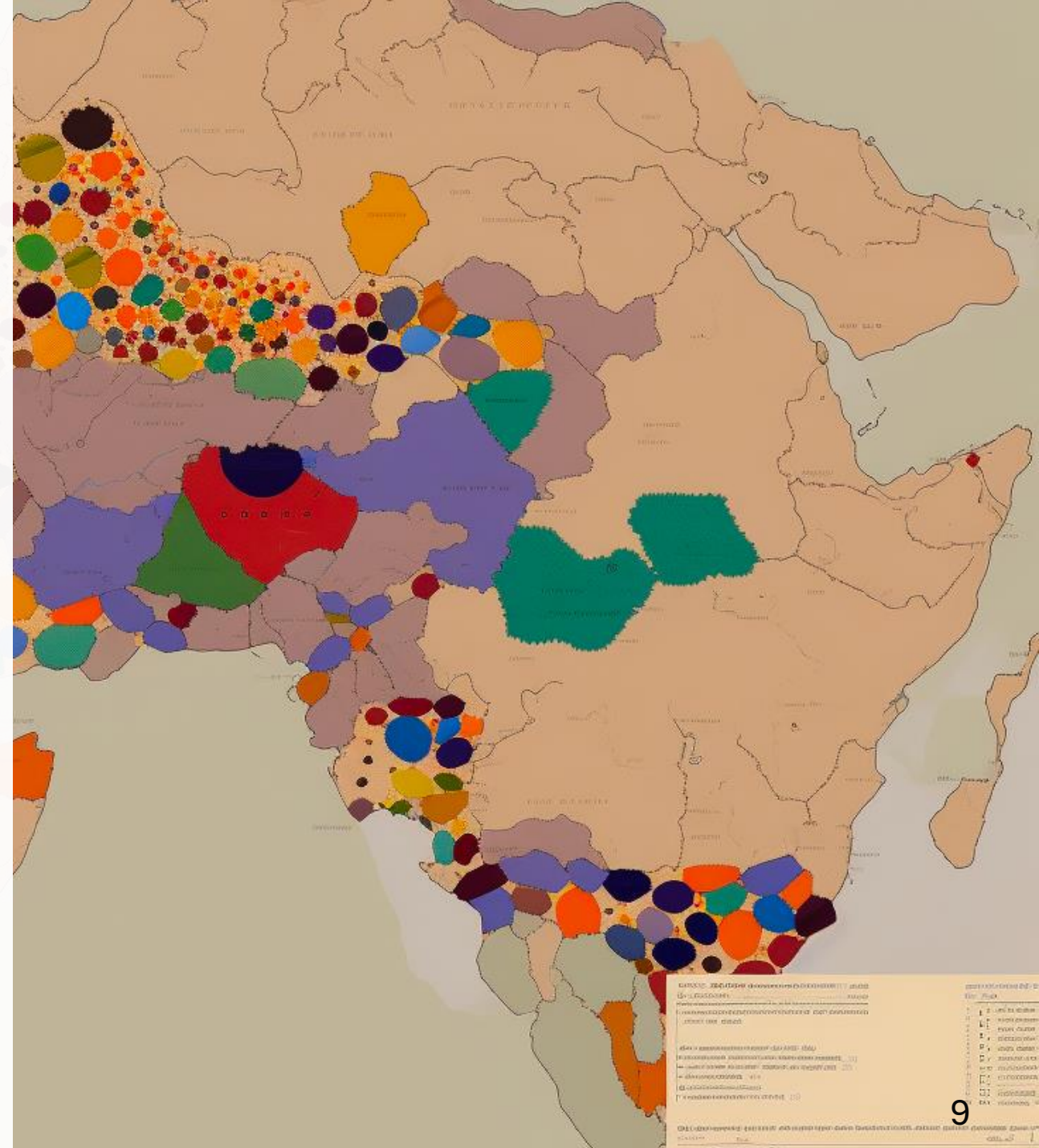
- Conflict dynamics arise from the environment
- Conflict onset and escalation are determined by:
  - Terrain constraints e.g. Infrastructure connectivity
  - Resources: availability + needs





### 3. Method

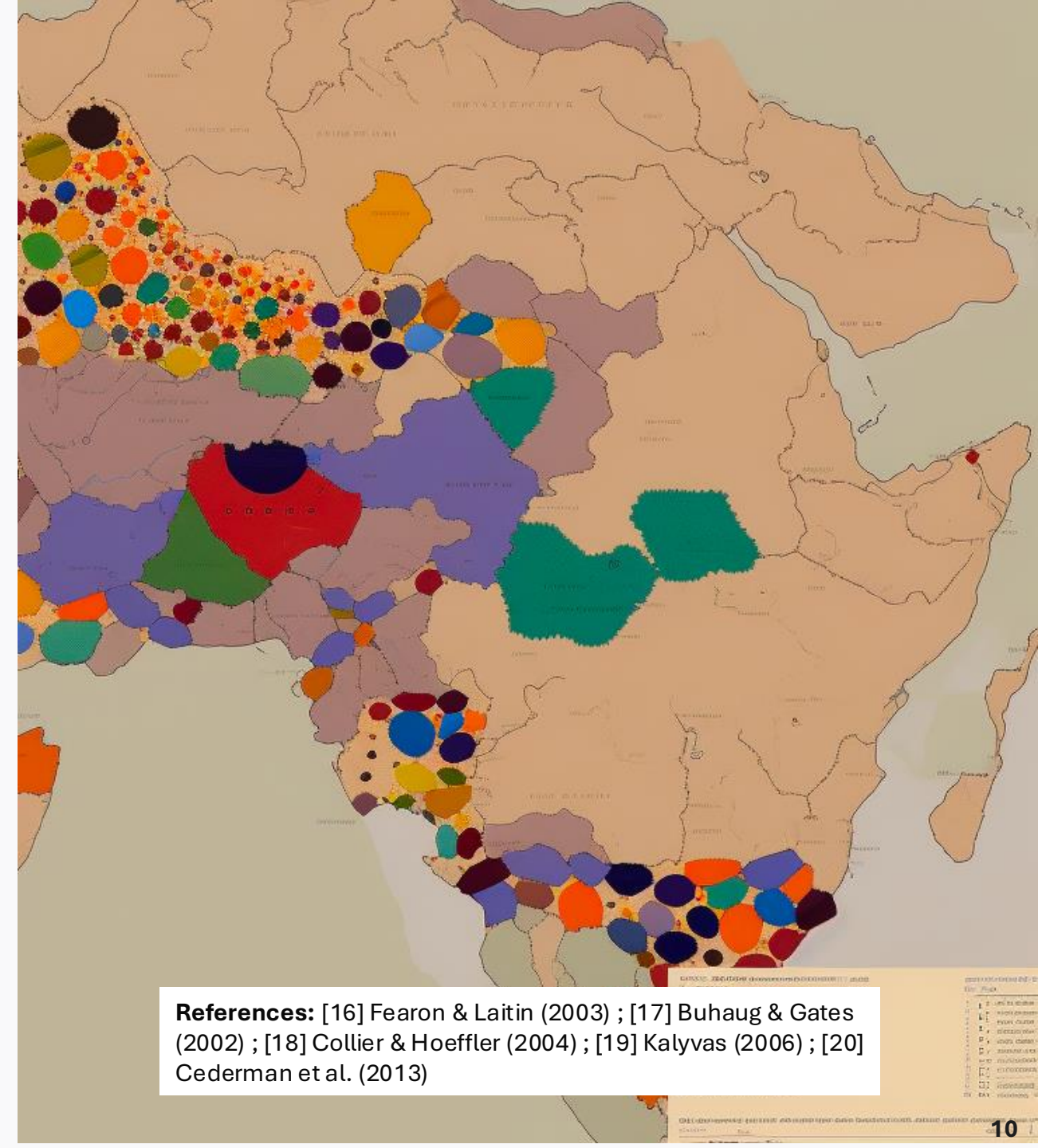
- We model the following aspects of evolution dynamics of conflict groups:
  1. Emergence
  2. Local Establishment
  3. Interaction
  4. Spatio-Temporal Diffusion



# 3.1 Emergence

*The way armed groups emerge is not uniformly distributed in space*

- The insurgence of armed group correlates with several human and geographical factors:
  - **Urban centres**
    - Cities offer resources and infrastructure to capture [18][17]
    - Urban areas serve as political organization hubs [19]
  - **Inhabited zones**
    - Conflict correlates with population presence and concentration [16][17].
    - State control is weaker in rural and peripheral regions [16][19]



**References:** [16] Fearon & Laitin (2003) ; [17] Buhaug & Gates (2002) ; [18] Collier & Hoeffler (2004) ; [19] Kalyvas (2006) ; [20] Cederman et al. (2013)



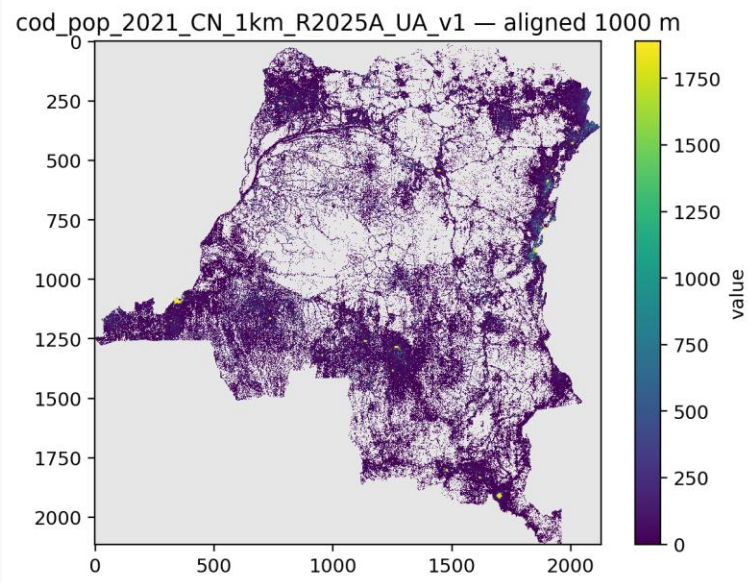
## 3.1 Emergence

*The way armed groups emerge is not uniformly distributed in space*

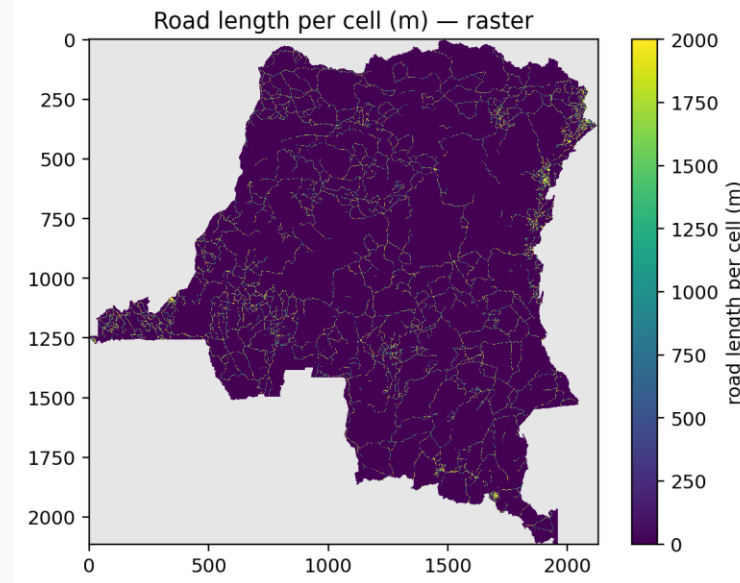
- **WorldPop** [21] is a public data service that provides population information in the form of TIFF files for countries worldwide.
- TIFF file: raster data made of pixels arranged in a grid.
  - Each pixel stores a numeric value (e.g., estimated population for that spatial cell).
  - Georeferenced so each pixel corresponds to a precise location in a coordinate system.

## 3.1 Emergence

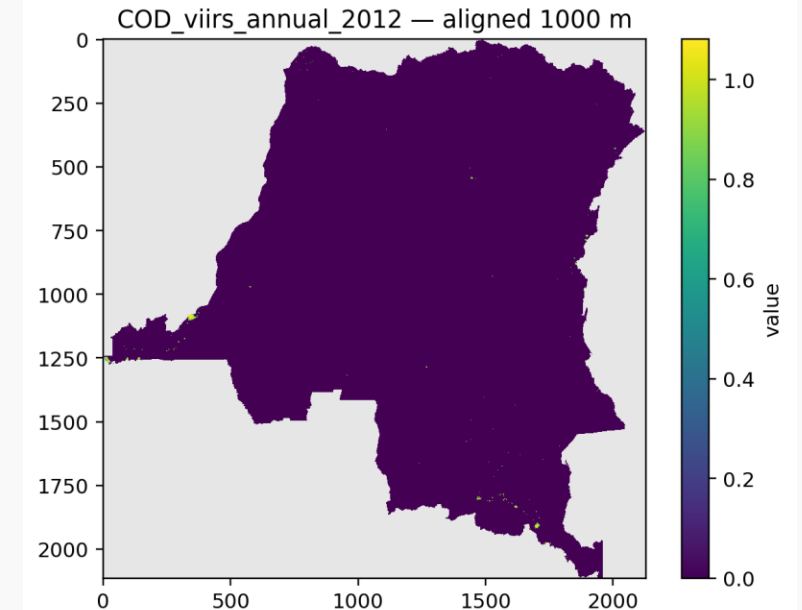
*The way armed groups emerge is not uniformly distributed in space*



Population



Roads



Nightlights



## 3.1 Emergence

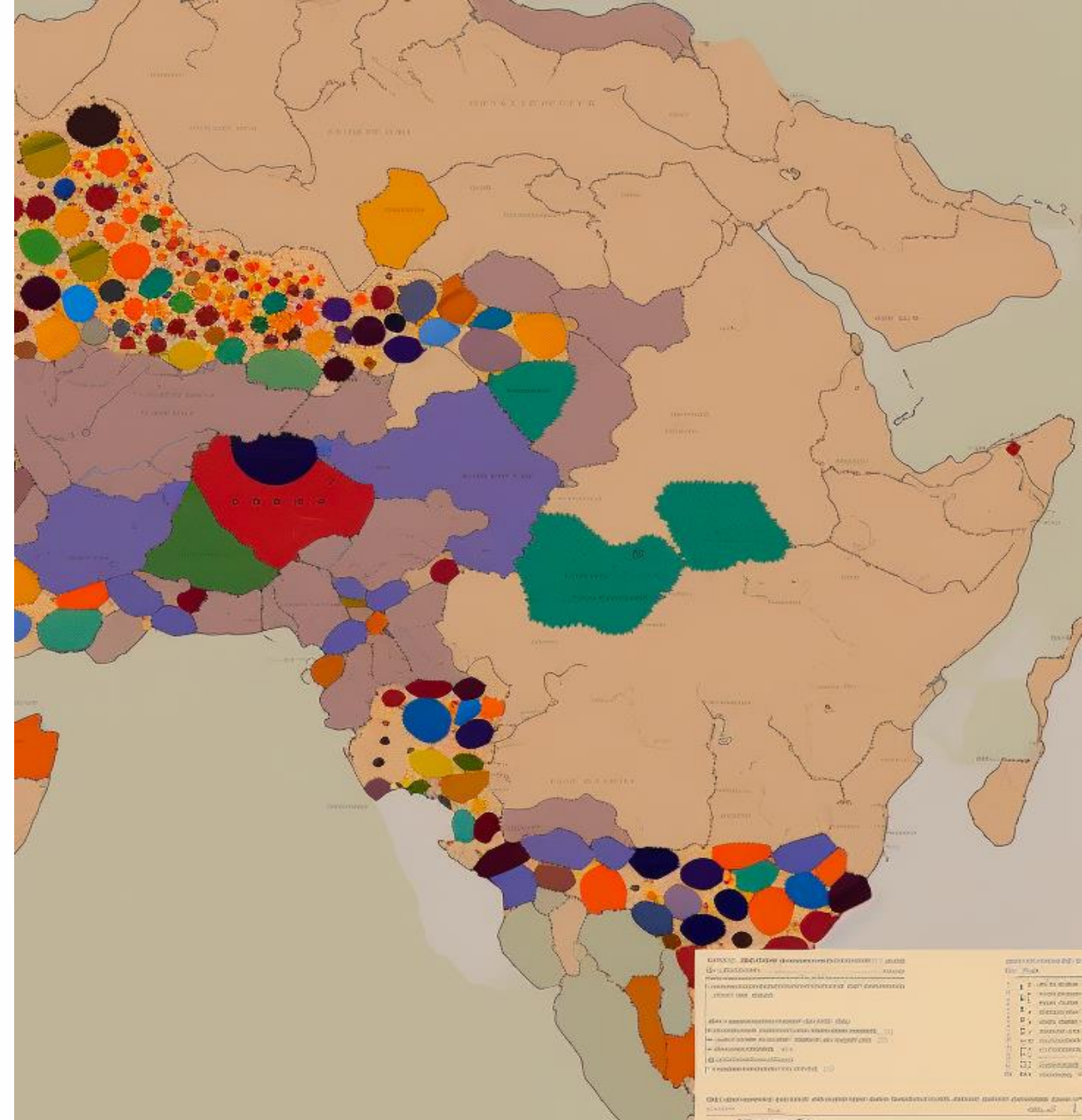
*The way armed groups emerge is not uniformly distributed in space*

- We define a probability distribution over our grid  $\mathcal{L}$ , proportional to our geo-features

$$U(\mathbf{x}) = \frac{1}{|\mathcal{F}|} \sum_{k \in \mathcal{F}} \Phi^k(\mathbf{x})$$

$$P_{\text{onset}}(\mathbf{x}) = \frac{\exp(U(\mathbf{x}))}{\sum_{\mathbf{z} \in \mathcal{L}} \exp(U(\mathbf{z}))}$$

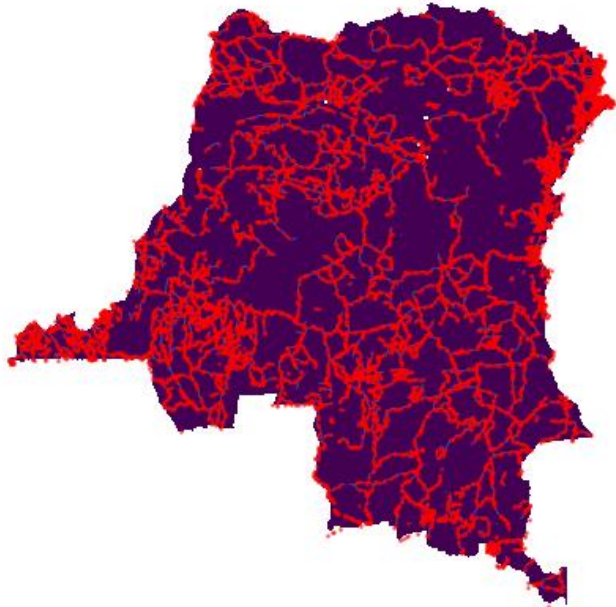
**References:** [16] Fearon & Laitin (2003) ; [17] Buhaug & Gates (2002) ; [18] Collier & Hoeffler (2004) ; [19] Kalyvas (2006) ; [20] Cederman et al. (2013)



## 3.1 Emergence

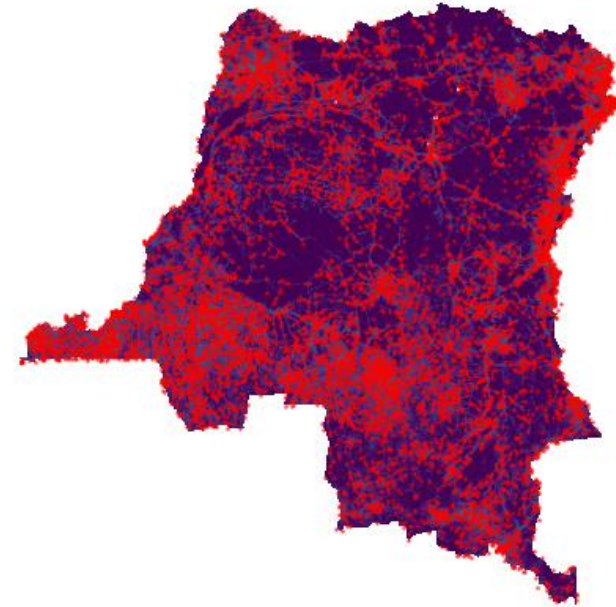
*The way armed groups emerge is not uniformly distributed in space*

Sample Map: roads



$$\Phi^{\text{roads}}(\mathbf{x})$$

Sample Map: pop



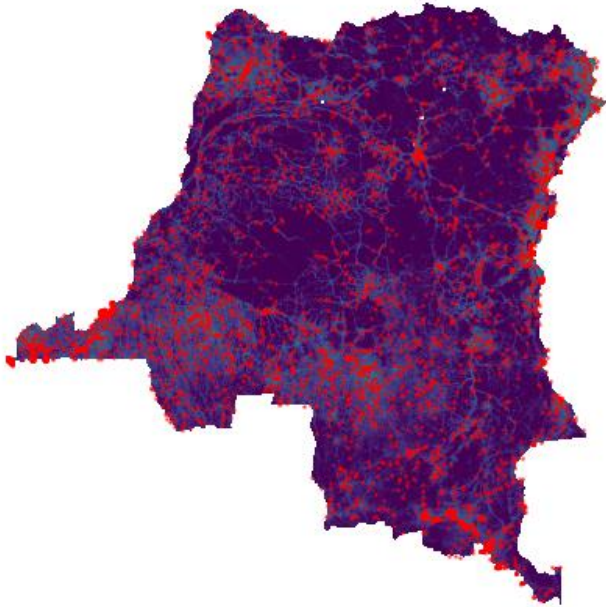
$$\Phi^{\text{pop}}(\mathbf{x})$$



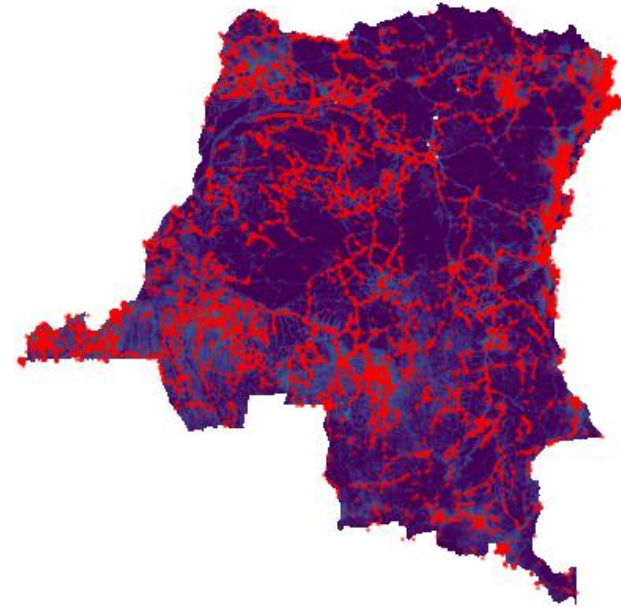
## 3.1 Emergence

*The way armed groups emerge is not uniformly distributed in space*

Sample Map: pop+viirs



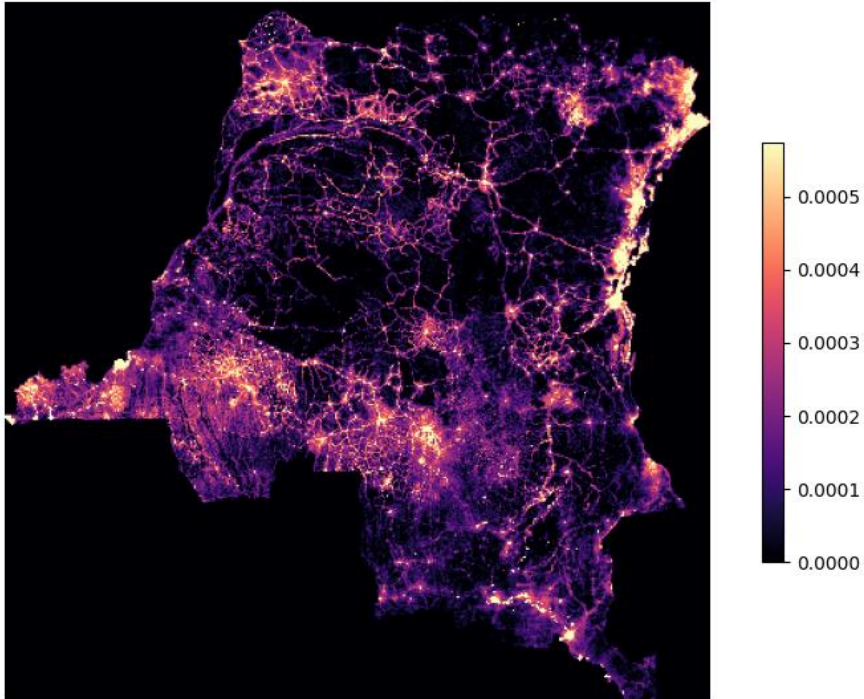
Sample Map: pop+roads



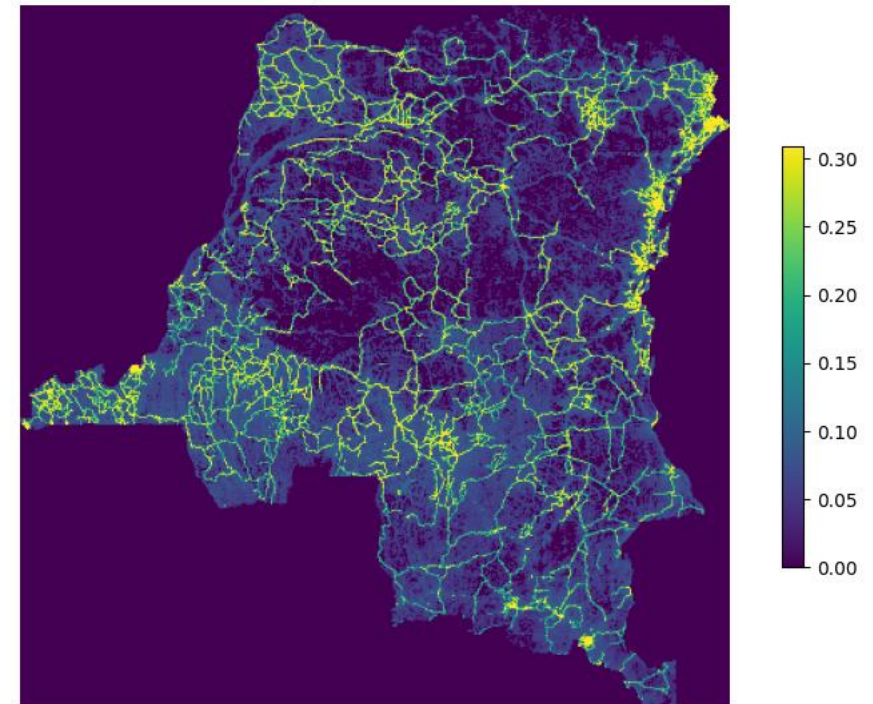
## 3.1 Emergence

*The way armed groups emerge is not uniformly distributed in space*

DOWN Start (Factor 5)  
Shape: (422, 425)



DOWN Move (Factor 5)  
Vis Gamma=0.3  
Shape: (422, 425)



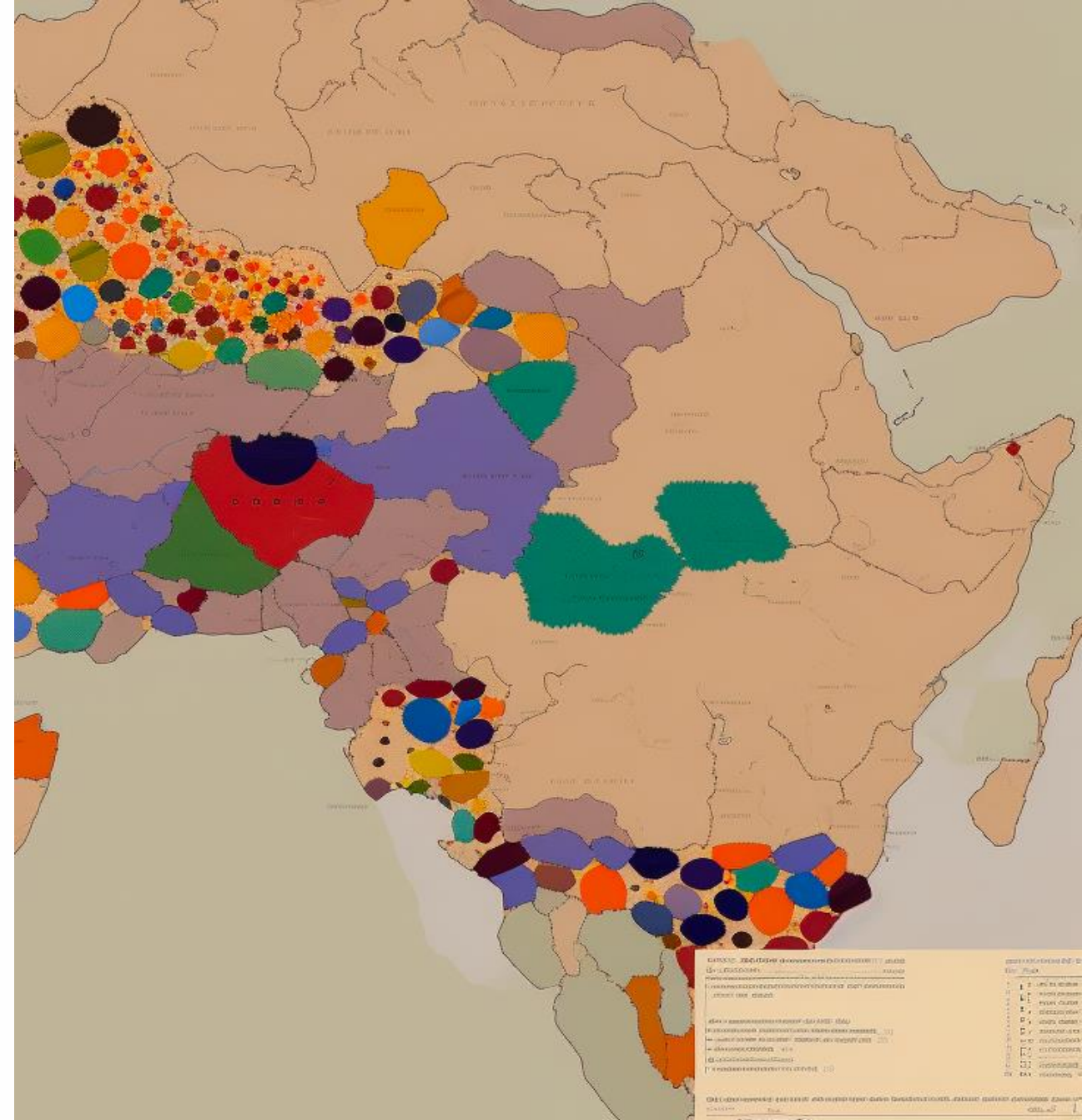


# 3.1 Emergence

*The way armed groups emerge is not uniformly distributed in space*

- Number of armed groups  $N$
- $\mathcal{G} = \{1, \dots, N\}$ .
- Spatial grid  $\mathcal{L}$
- $t \in \mathcal{T} = \{0, 1, \dots, T_{\max}\}$

$(t = 0) \quad P_{\text{onset}}(\mathbf{x}) \rightarrow \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$



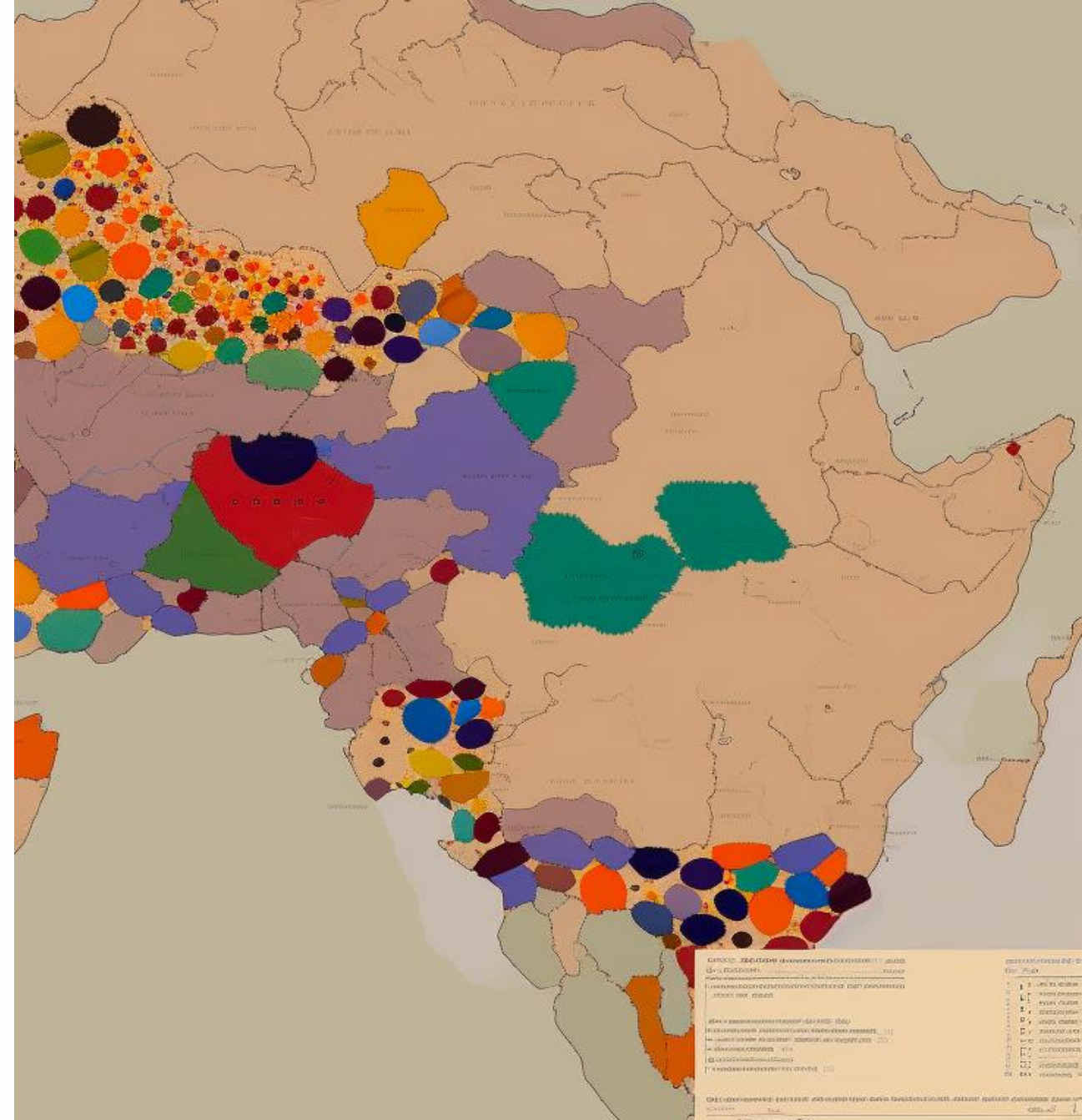
# 3.1 Emergence

*The way armed groups emerge is not uniformly distributed in space*

$$(t = 0) \quad P_{\text{onset}}(\mathbf{x}) \rightarrow \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$$

→ Influence of the group denoted as  $S_g$

$$S_g(\mathbf{x}, 0) = \begin{cases} S_{\text{init}} & \text{if } \mathbf{x} = \mathbf{x}_g \\ 0 & \text{otherwise} \end{cases}$$





## 3.2 Establishment

*The way armed groups grow depends on their available resources*

- Resources of cell  $x$

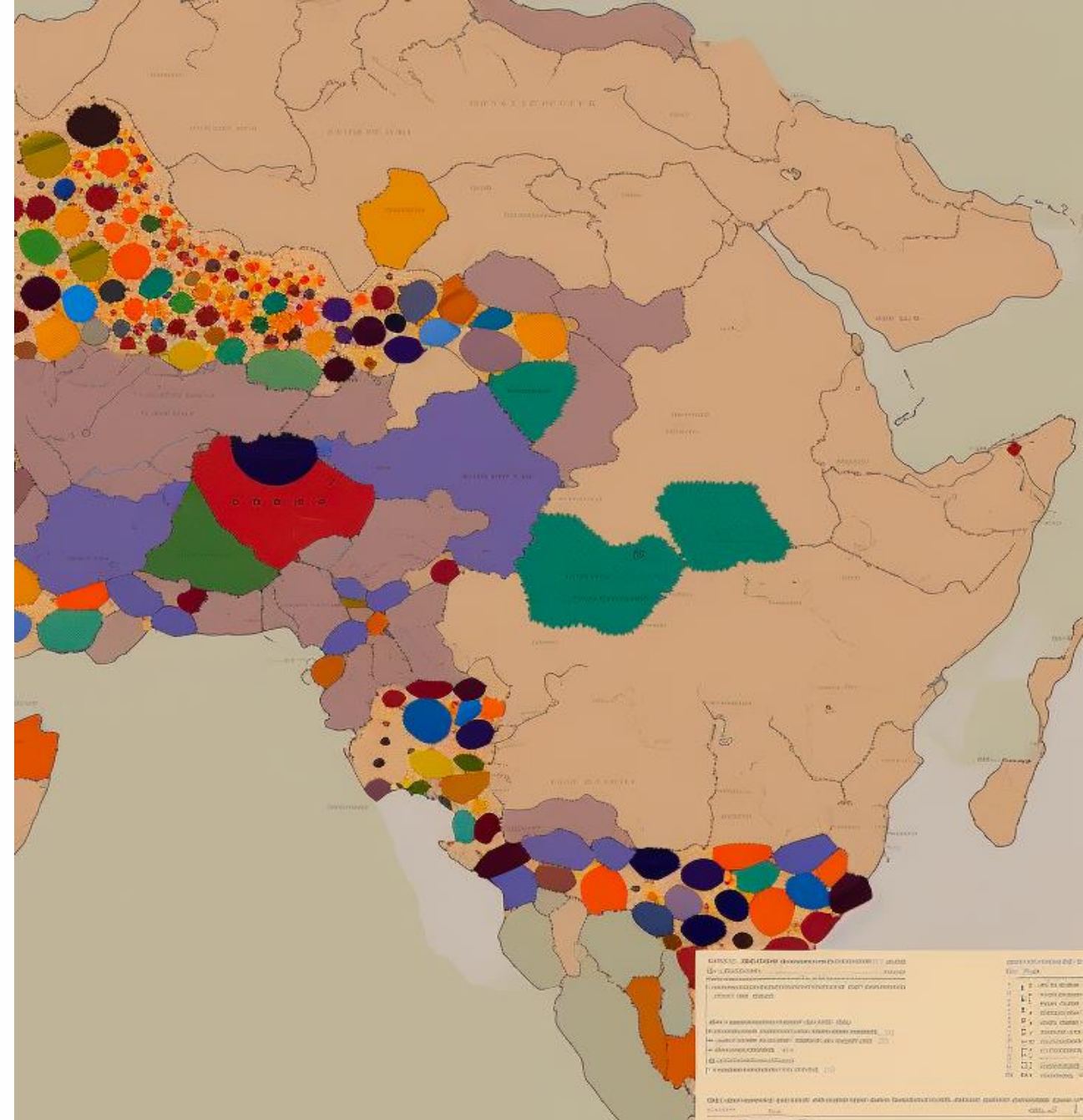
$$K(\mathbf{x}) = M \cdot \left( \frac{\exp(\Psi(\mathbf{x}))}{\max_{\mathbf{z} \in \mathcal{L}} [\exp(\Psi(\mathbf{z}))]} \right)^\gamma$$

$$\Psi(\mathbf{x}) = \Phi^{\text{roads}}(\mathbf{x}) + \Phi^{\text{pop}}(\mathbf{x})$$

- Growth rate

$$r(\mathbf{x}) = R \cdot K(\mathbf{x})$$

We also explore modeling **resource finiteness** [\*]



## 3.2 Establishment

- Updated Influence

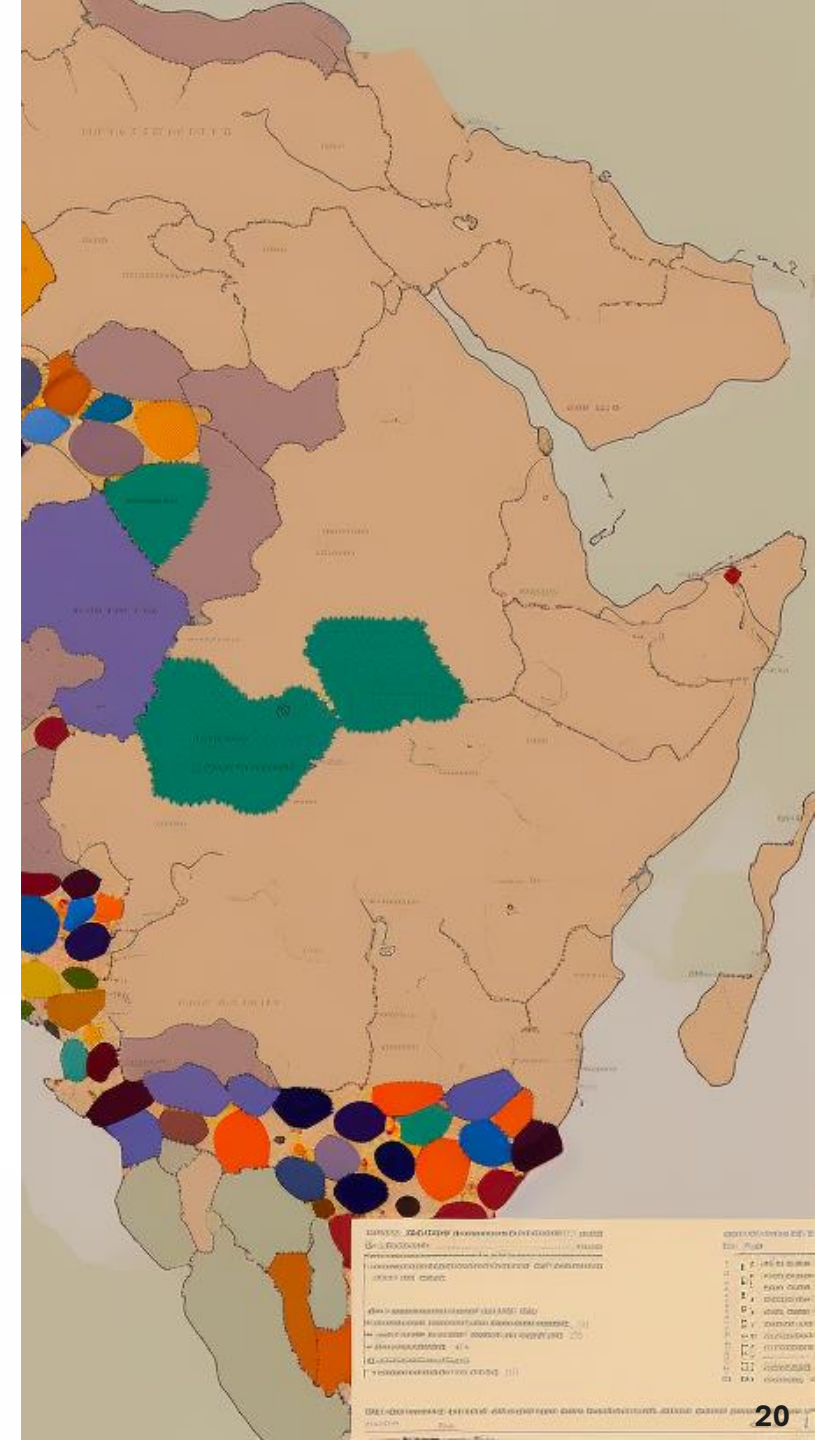
$$\tilde{S}_g(\mathbf{x}, t) = S_g(\mathbf{x}, t) \cdot (1 + r(\mathbf{x}) + \Pi_g(t))$$

**Retained Mass**  
(what stays)

$$S_g^{\text{ret}}(\mathbf{x}, t) = \min \left( \tilde{S}_g(\mathbf{x}, t), K(\mathbf{x}) \right)$$

**Excess Mass**  
(What moves)

$$S_g^{\text{exc}}(\mathbf{x}, t) = \max \left( 0, \tilde{S}_g(\mathbf{x}, t) - K(\mathbf{x}) \right)$$





### 3.3 Interaction

*The way armed groups grow / shrink depends on how they interact*

- **When and where does an interaction occur?**

- *Spatial Area of Influence of group  $g$*

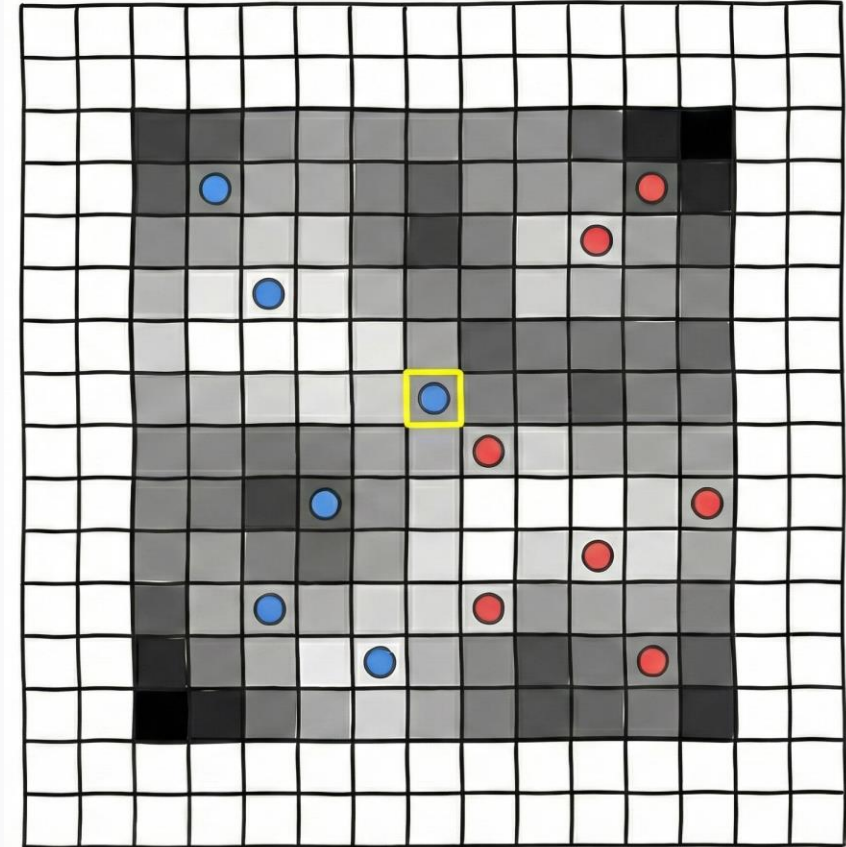
$$\Omega_g(t) = \{\mathbf{x} \in \mathcal{L} \mid S_g(\mathbf{x}, t) > \epsilon\}$$

- *One-pixel expansion of the above*

$$\Omega_g^\oplus(t) = \Omega_g(t) \oplus B = \{\mathbf{x} + \mathbf{b} \mid \mathbf{x} \in \Omega_g(t), \mathbf{b} \in B\}$$

- *Indicator function of interaction between group  $j$  and  $i$*

$$\delta_{ij}(t) = 1 \iff |\Omega_i(t) \cap \Omega_j^\oplus(t)| > 0$$



### 3.3 Interaction

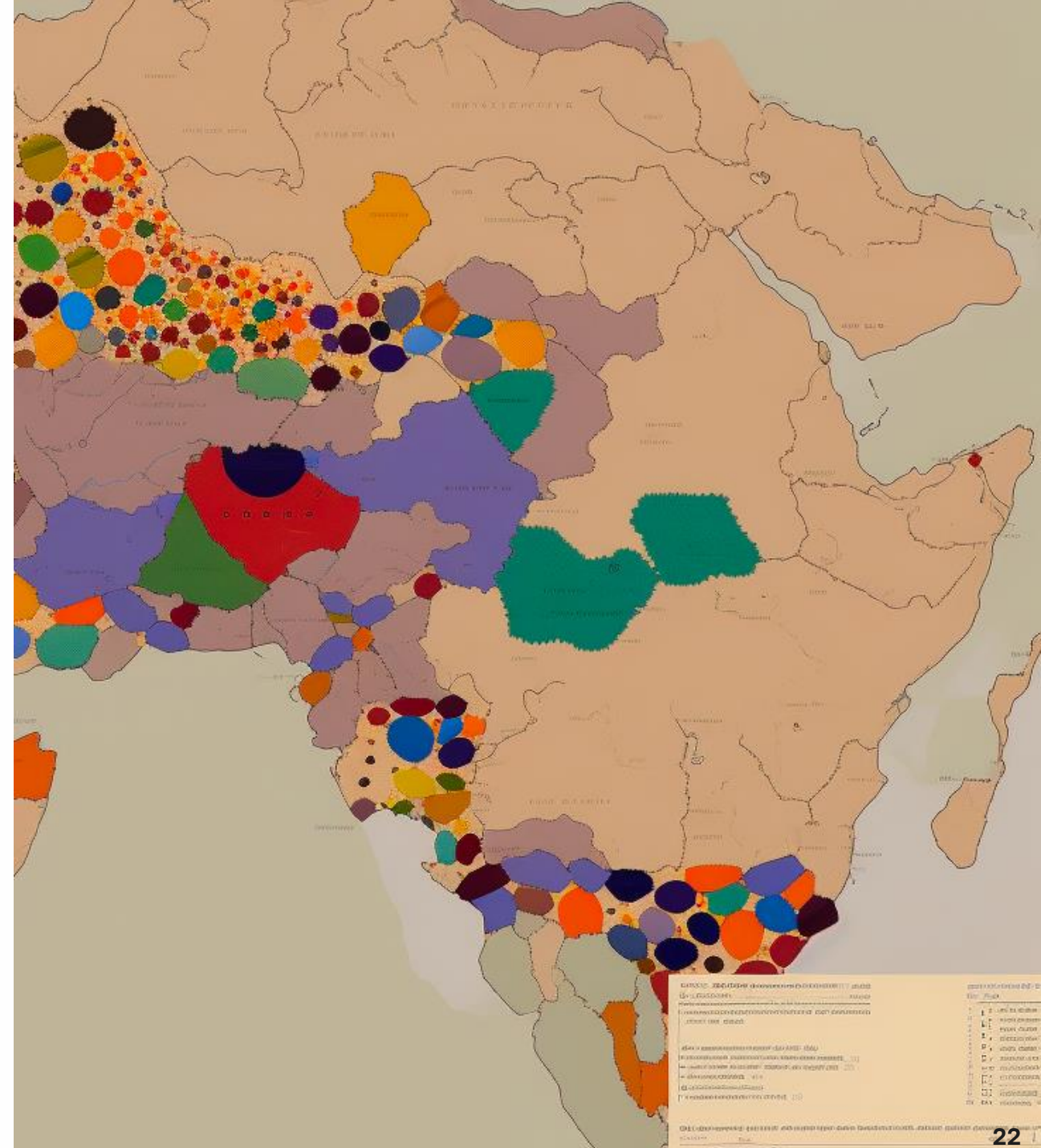
*The way armed groups grow / shrink depends on how they interact*

- Does a group cooperate or fight?

$$a_{ij} \in \{\text{Cooperate (C), Defect (D)}\}$$

| Group $i \setminus$ Group $j$ | Cooperate ( $a = 1$ ) | Defect ( $a = 0$ ) |
|-------------------------------|-----------------------|--------------------|
| Cooperate ( $a = 1$ )         | $R$                   | $S$                |
| Defect ( $a = 0$ )            | $T$                   | $P$                |

→ We choose values such that  $T > R > P > S$

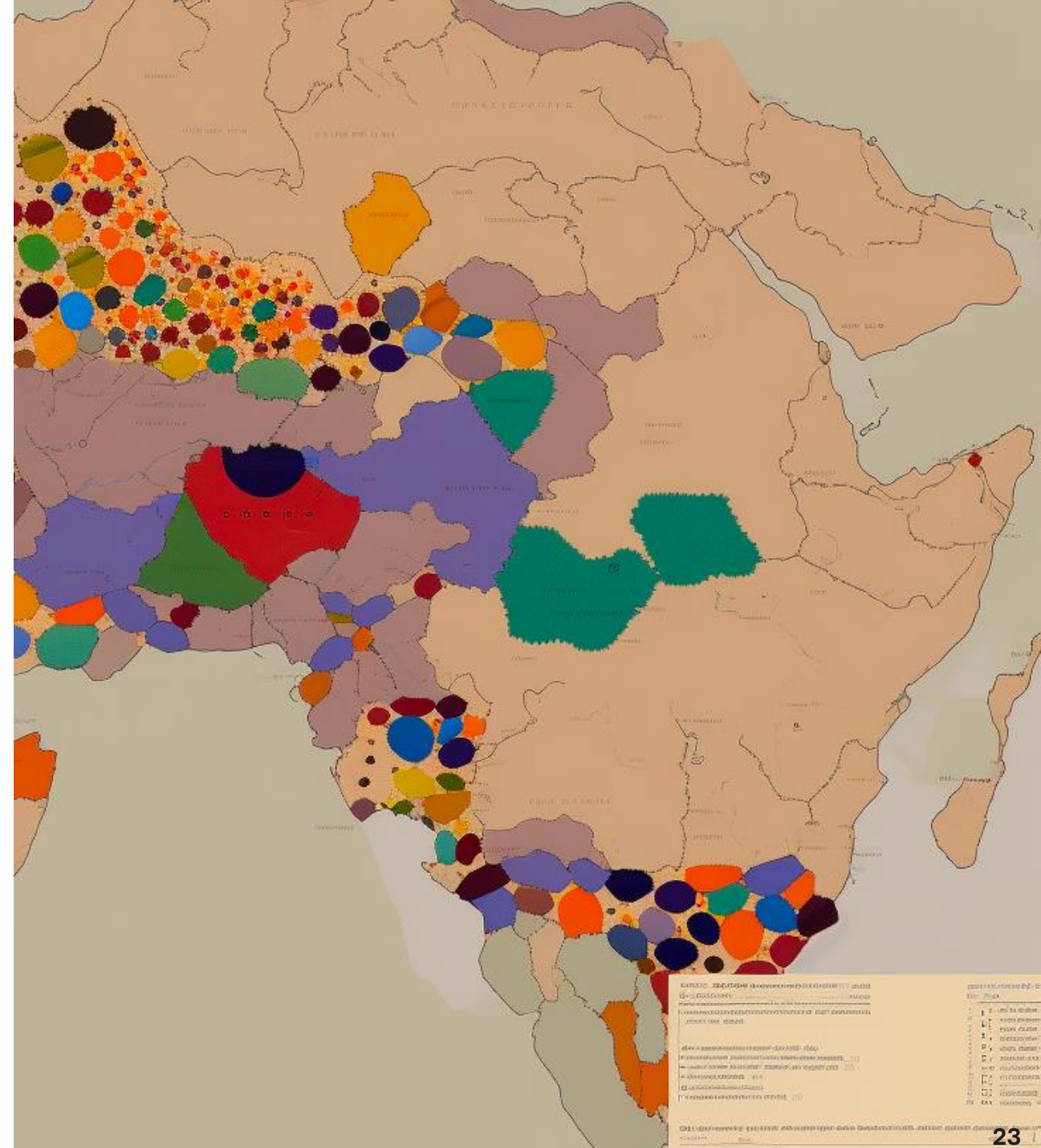




### 3.3 Interaction

*The way armed groups grow / shrink depends on how they interact*

- We define 4 different strategies:
  - Always **Defect (D)**
  - Always **Cooperate (C)**
  - Random **(R)**
  - Tit-for-Tat **(T)**
- We also explore **pure resource maximization** strategies [\*]

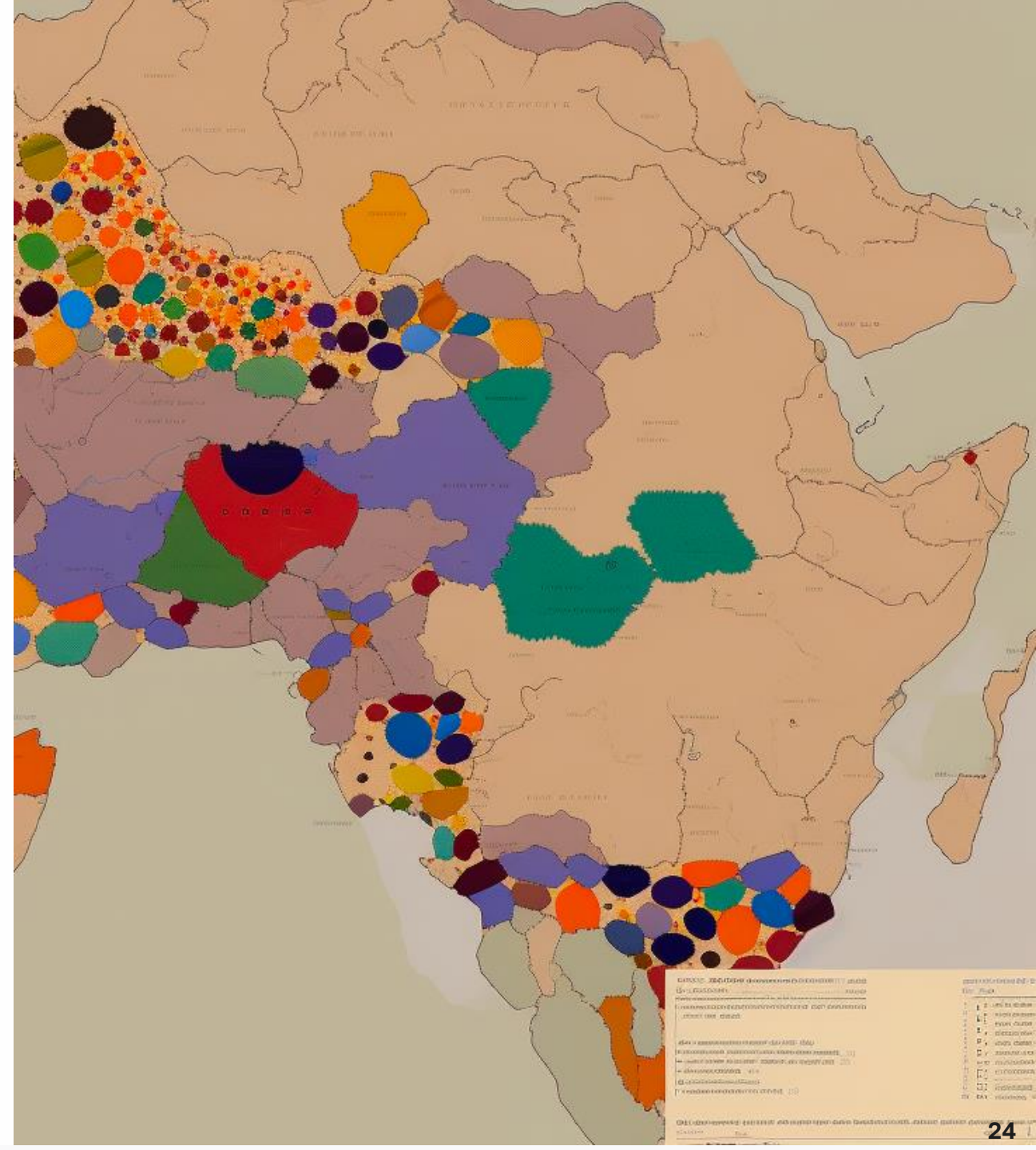


### 3.3 Interaction

*The way armed groups grow / shrink depends on how they interact*

$$V_{ij} = \begin{cases} \underbrace{\left( \frac{S_j(t)}{S_i(t) + S_j(t)} \right) \cdot R}_{\text{Relative Size Scaling}} & \text{if } a_i = 1, a_j = 1 \quad (\text{Peace}) \\ \left( \frac{S_j(t)}{S_i(t) + S_j(t)} \right) \cdot P & \text{if } a_i = 0, a_j = 0 \quad (\text{War}) \\ S & \text{if } a_i = 1, a_j = 0 \quad (\text{Sucker}) \\ T & \text{if } a_i = 0, a_j = 1 \quad (\text{Predation}) \end{cases}$$

$$\Pi_g(t) = \sum_{j \neq g} V_{gj}(t)$$





## 3.4 Spatio-Temporal Diffusion

- The portion of influence that goes to neighbour  $\mathbf{x}'$

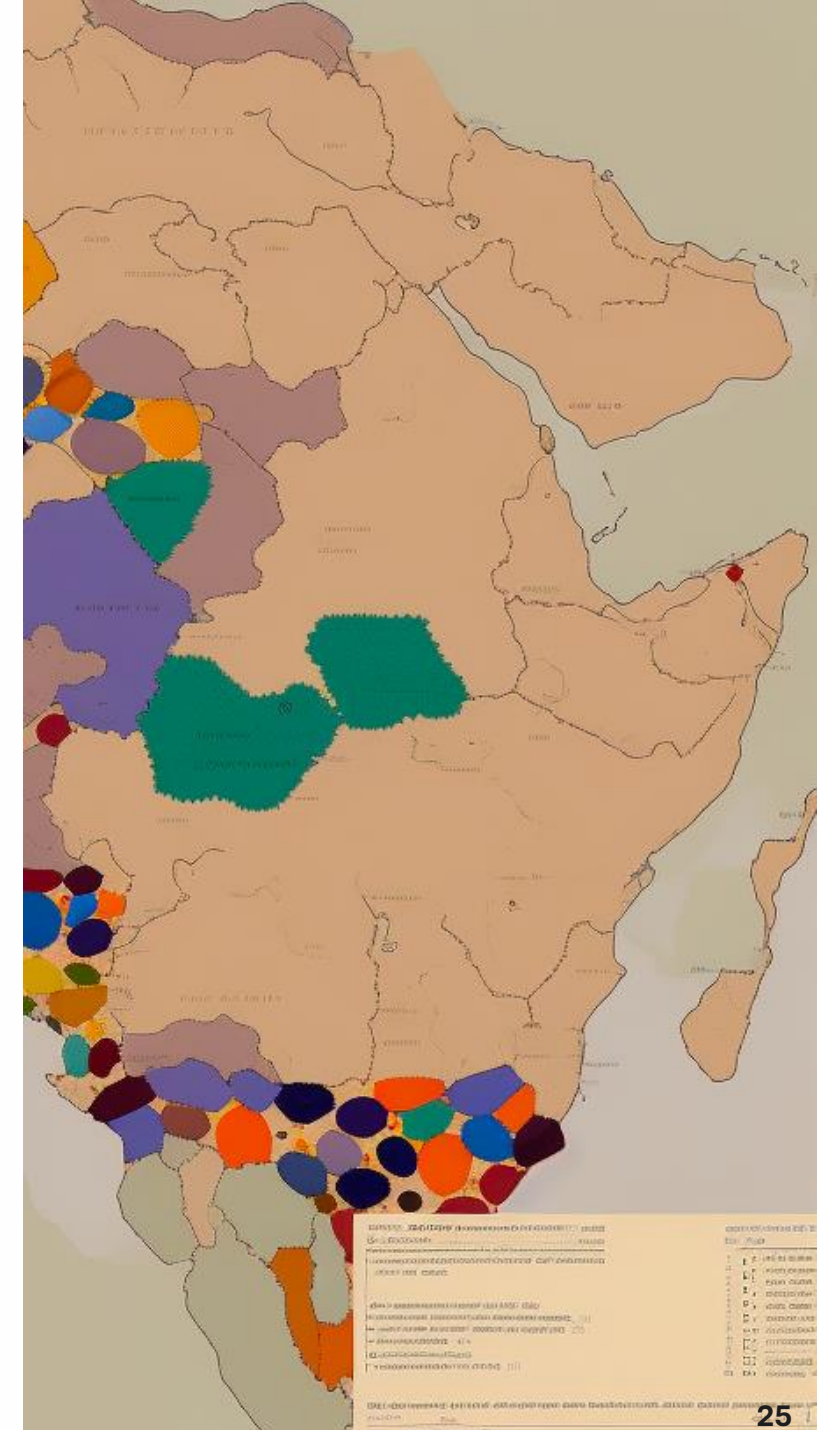
$$f_{\mathbf{x} \rightarrow \mathbf{x}'} = S_g^{\text{exc}}(\mathbf{x}, t) \cdot \frac{K_g^{\text{eff}}(\mathbf{x}')}{\sum_{\mathbf{n} \in \text{neighbors}} K_g^{\text{eff}}(\mathbf{n})}$$

$$K_g^{\text{eff}}(\mathbf{x}') = \begin{cases} 0 & \text{if } \mathbf{x}' \text{ is occupied by an enemy} \\ K(\mathbf{x}') & \text{otherwise} \end{cases}$$

**Update**

$$S_g(\mathbf{x}, t + 1) = S_g^{\text{ret}}(\mathbf{x}, t) + \sum_{\mathbf{x}' \in \mathcal{N}(\mathbf{x})} f_{\mathbf{x}' \rightarrow \mathbf{x}}$$

|      |        |       |
|------|--------|-------|
|      | TOP    |       |
| LEFT | STAY   | RIGHT |
|      | BOTTOM |       |

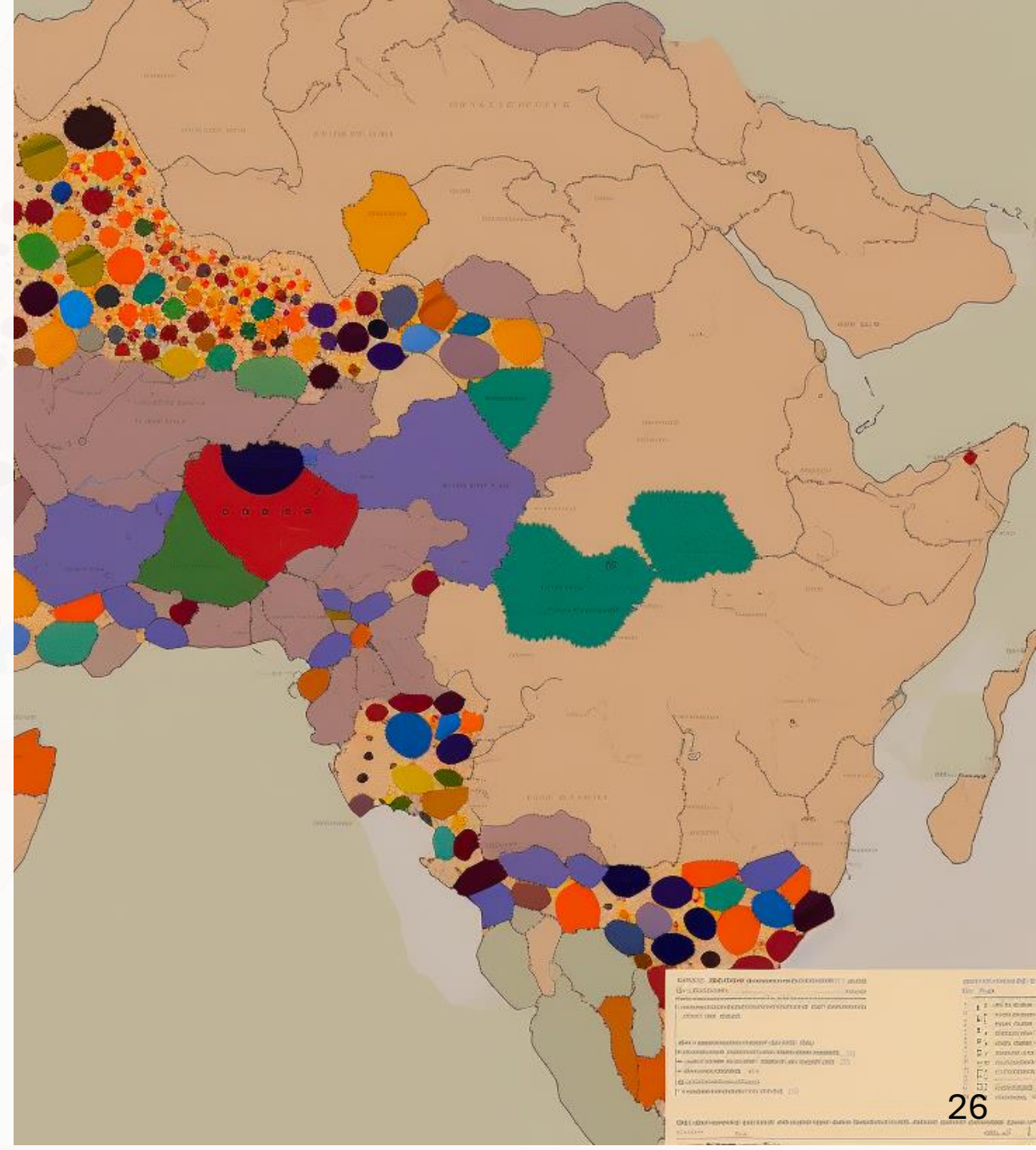


## 4. Analysis & Results

- *Analyze*: how do **different priors** affect the **initial spatial density** of groups

### *Quantitative Metrics:*

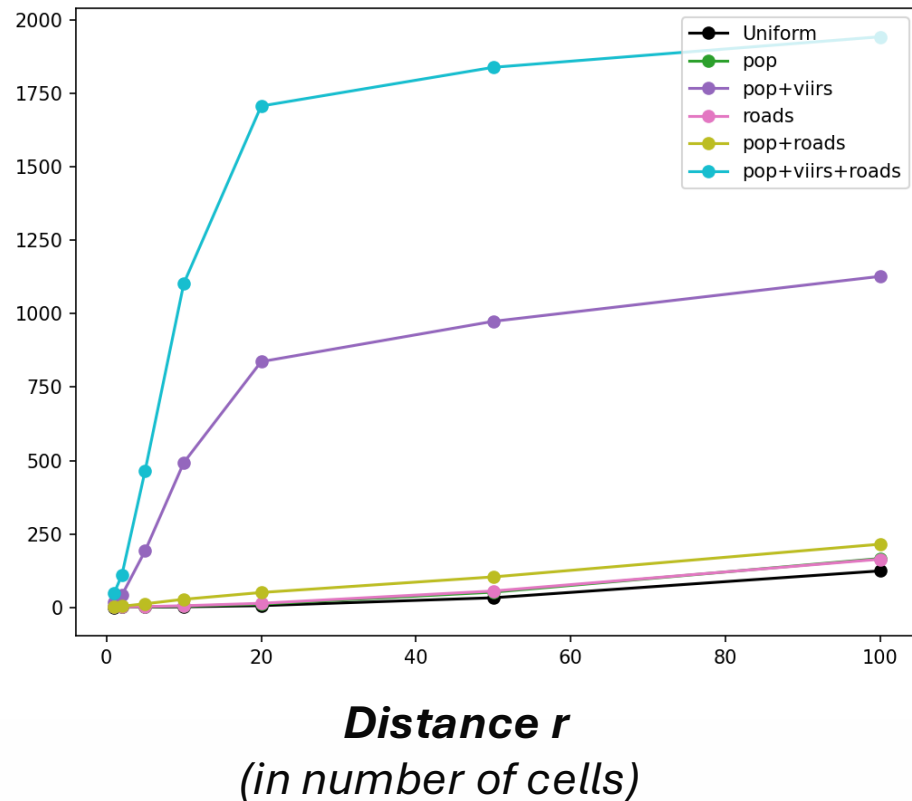
- For a group, **how many** other groups lie within distance  $r$ ?
  - We choose
$$r = [1, 2, 5, 10, 20, 50, 100]$$
- What's the distribution of the **nearest-neighbour** distance?



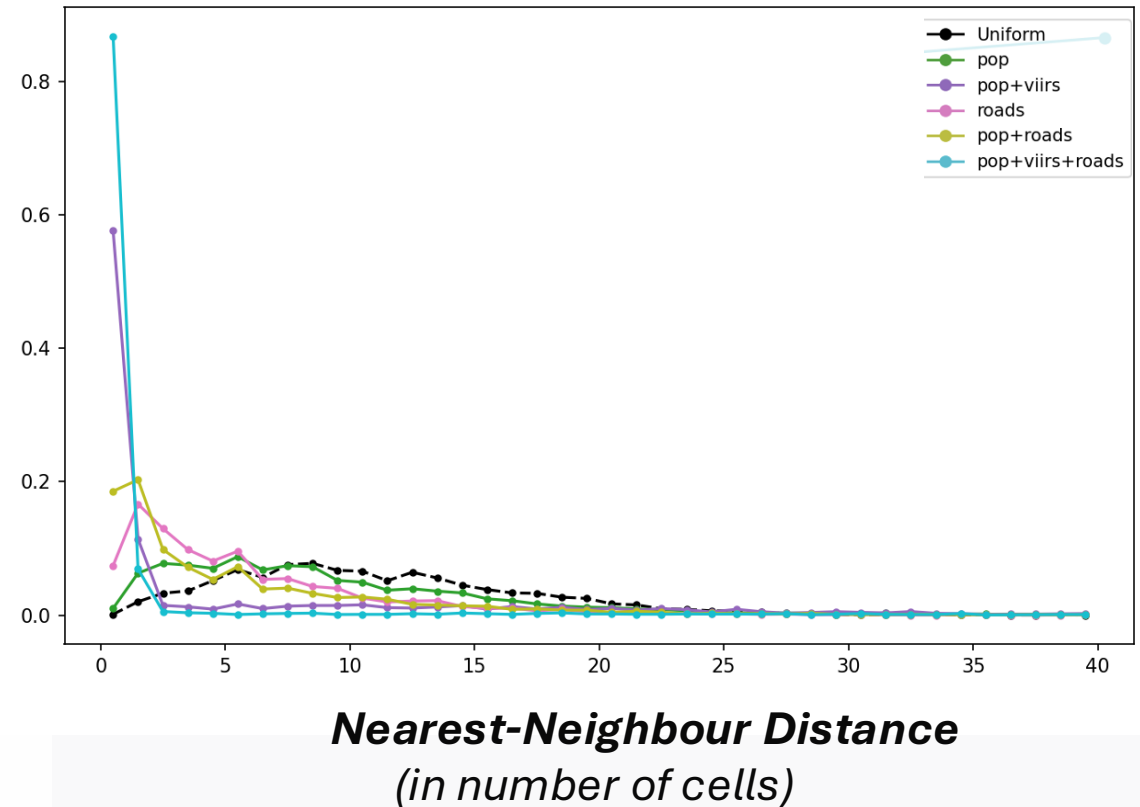


## 4.1. Sensitivity Analysis on Conflict Onset

*How many other groups lie within distance  $r$ ?*



*Nearest-Neighbour Density*



## 4.2 Sensitivity Analysis on Growth Rate

- *Growth Rate*      $r(\mathbf{x}) = R \cdot K(\mathbf{x})$

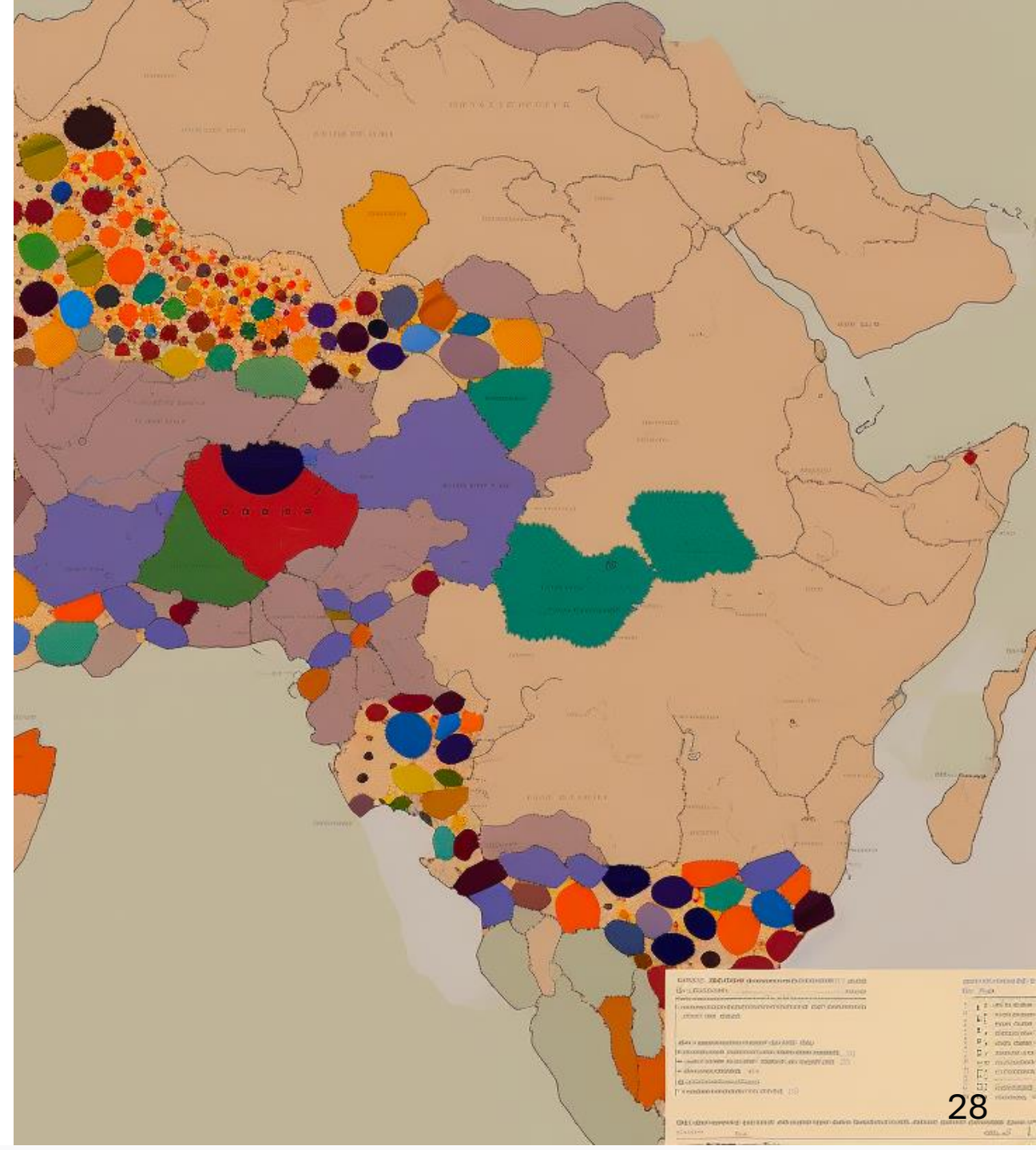
We ask:

- *How does the area of groups evolve over time?*
- *How does that change among groups with different strategies?*

**Groups: 10**

**Strategy Distribution: Uniform**

**Repetitions: 10**

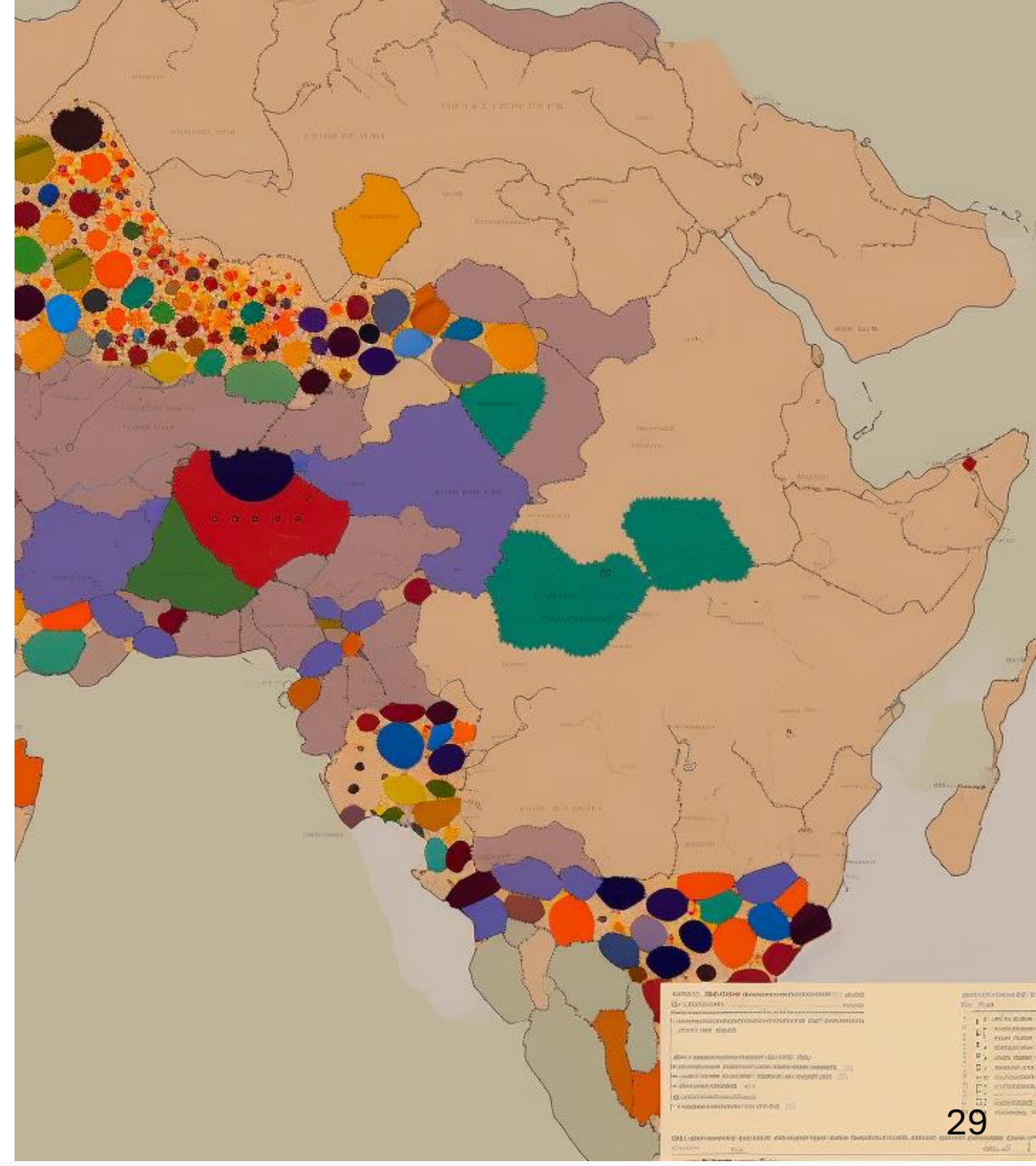




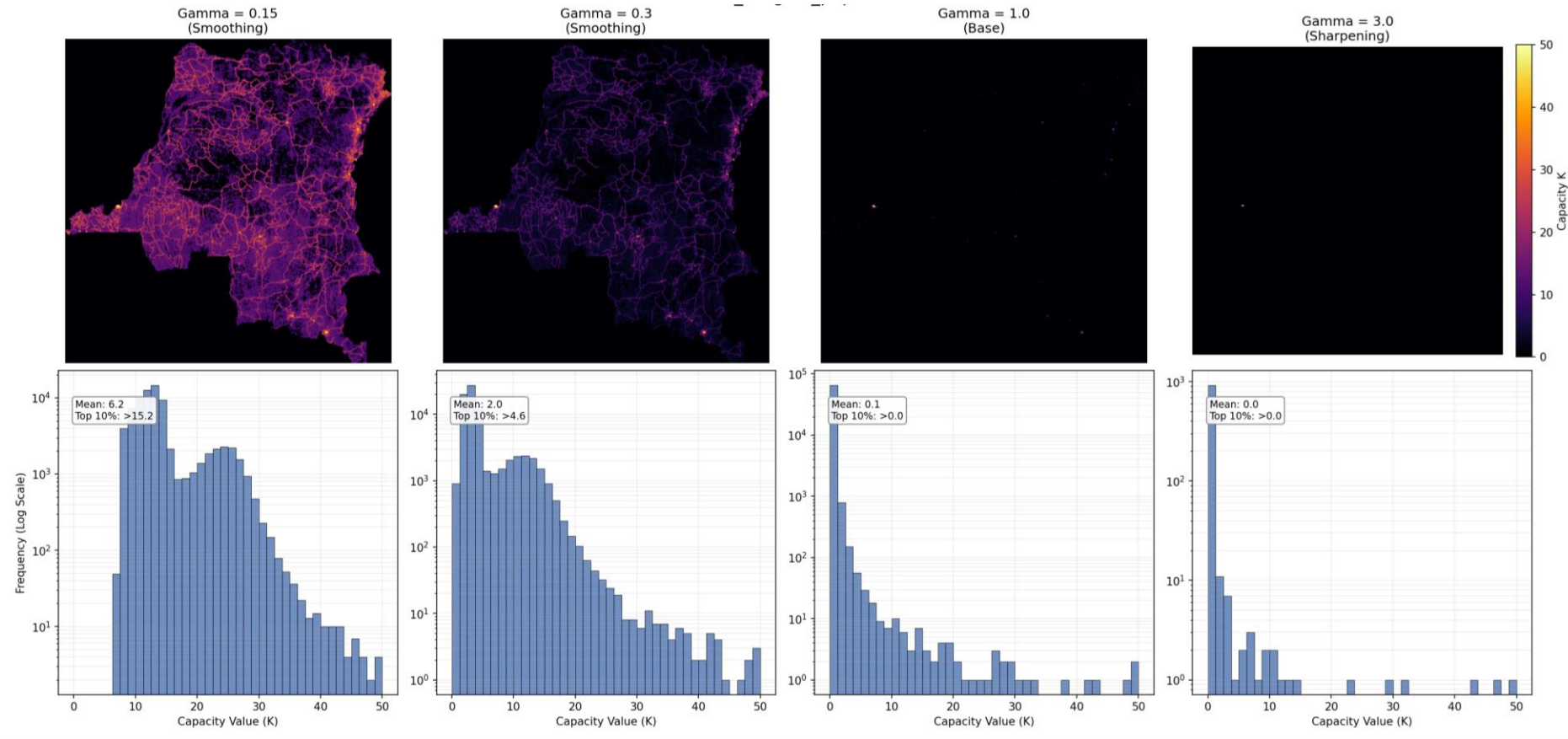
## 4.2 Calibration on Smoothing Parameter Gamma

$$K(\mathbf{x}) = M \cdot \left( \frac{\exp(\Psi(\mathbf{x}))}{\max_{\mathbf{z} \in \mathcal{L}} [\exp(\Psi(\mathbf{z}))]} \right)^\gamma$$

- We qualitatively inspect the distribution to ensure resources are not too skewed,
- necessary **calibration phase**.

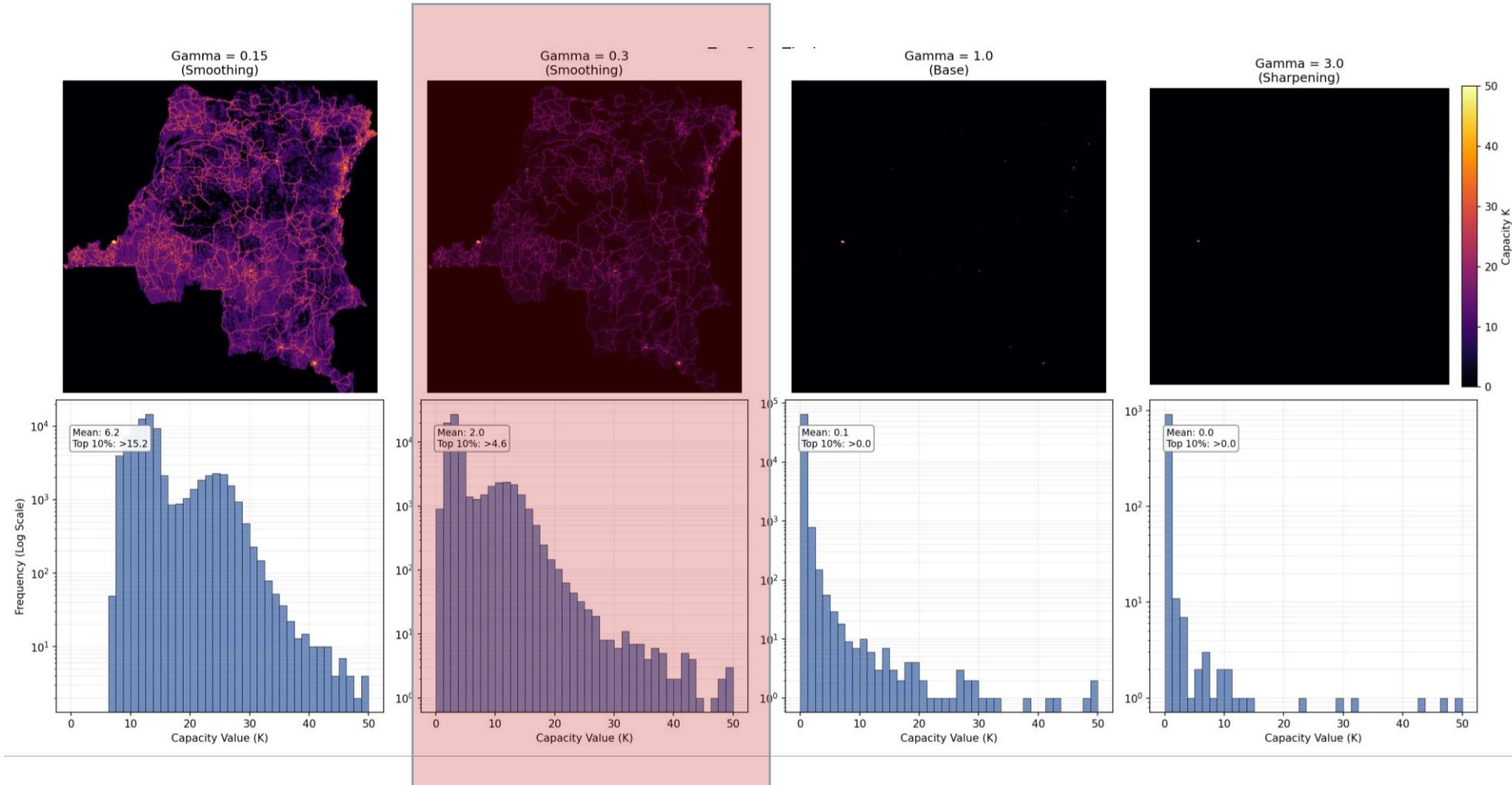


## 4.2 Sensitivity Analysis on Smoothing Parameter Gamma



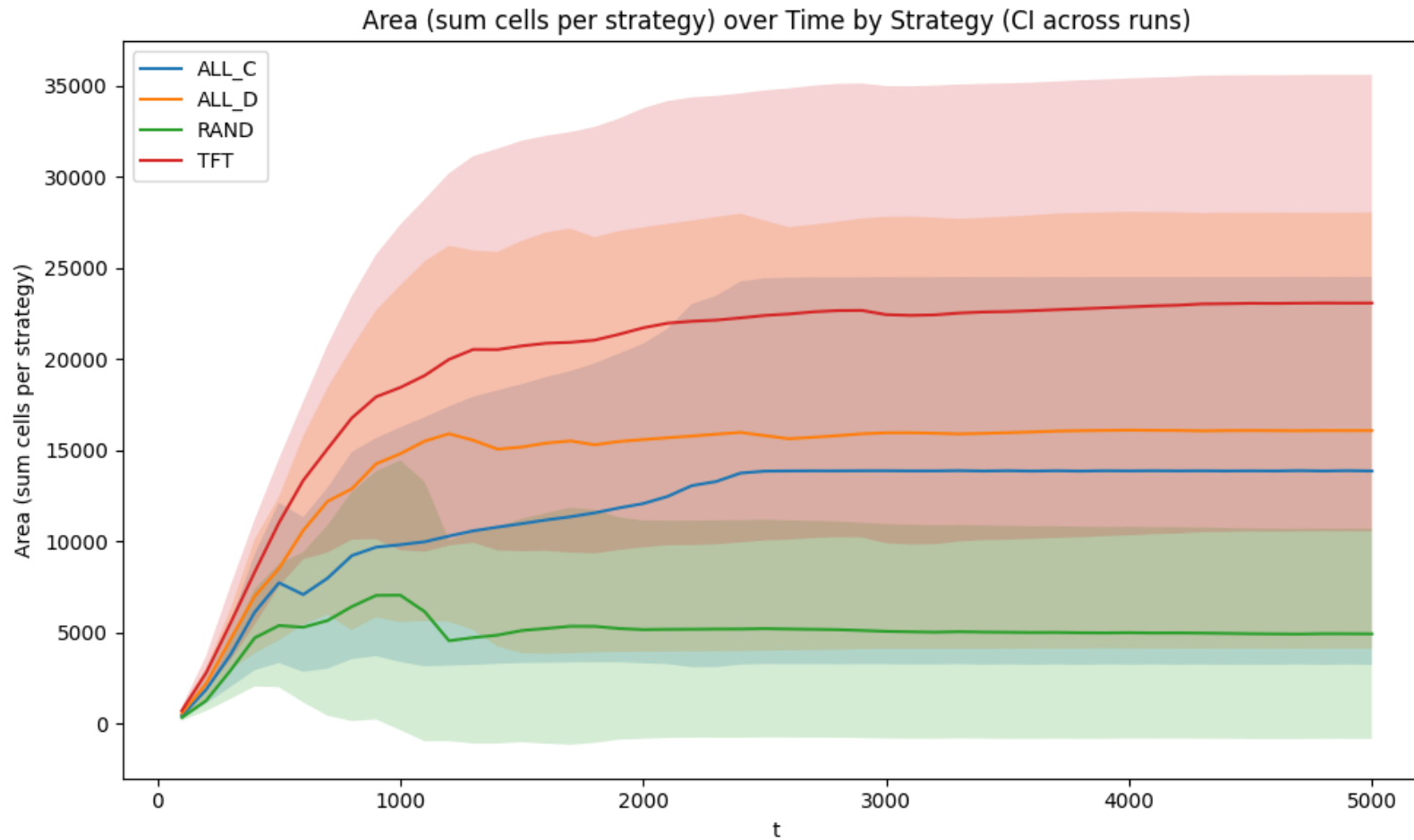


## 4.2 Sensitivity Analysis on Smoothing Parameter Gamma



## 4.2 Sensitivity Analysis on Growth Rate

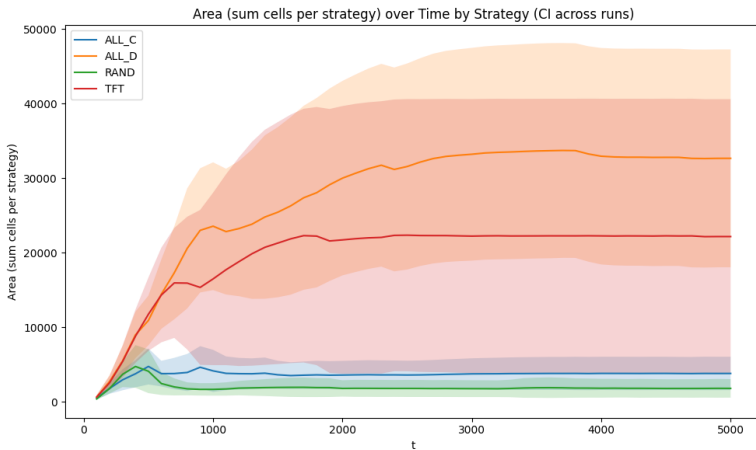
**$R = 0.60$**



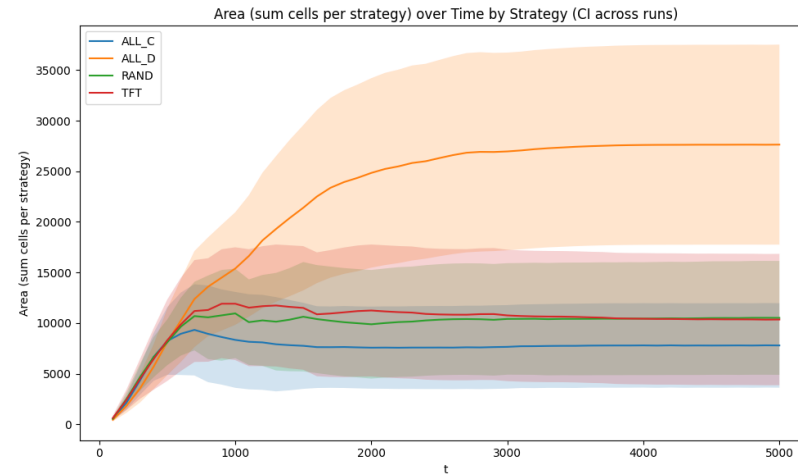


## 4.2 Sensitivity Analysis on Growth Rate

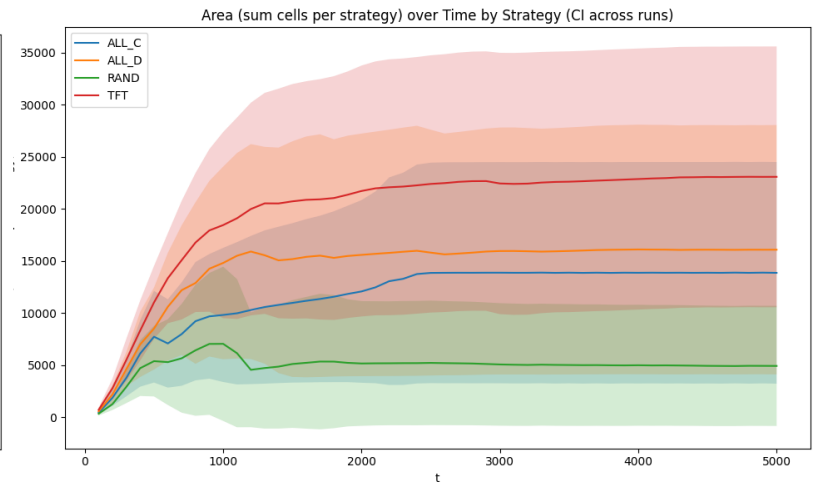
$R = 0.10$



$R = 0.30$



$R = 0.60$



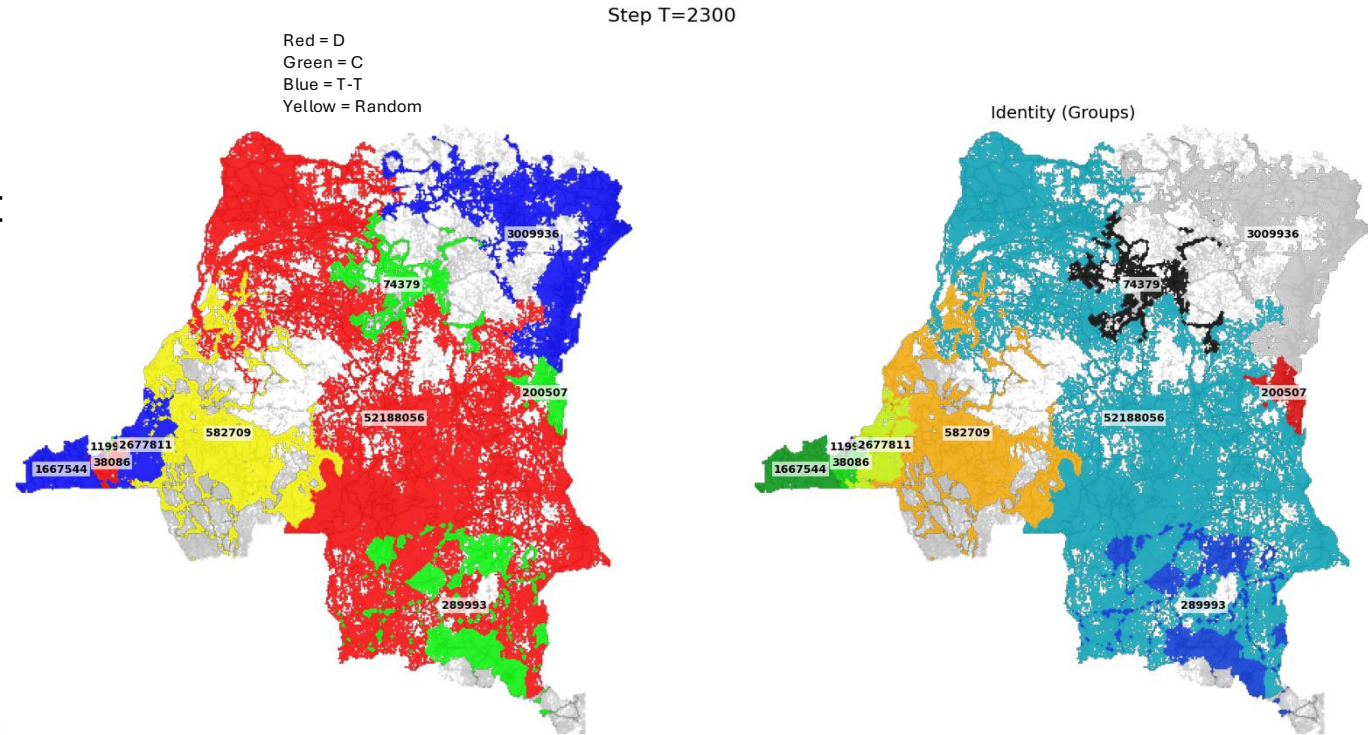
Results are aggregated across multiple runs, with:

- **Solid lines** = average area per strategy
- **Shaded regions** = variance over runs

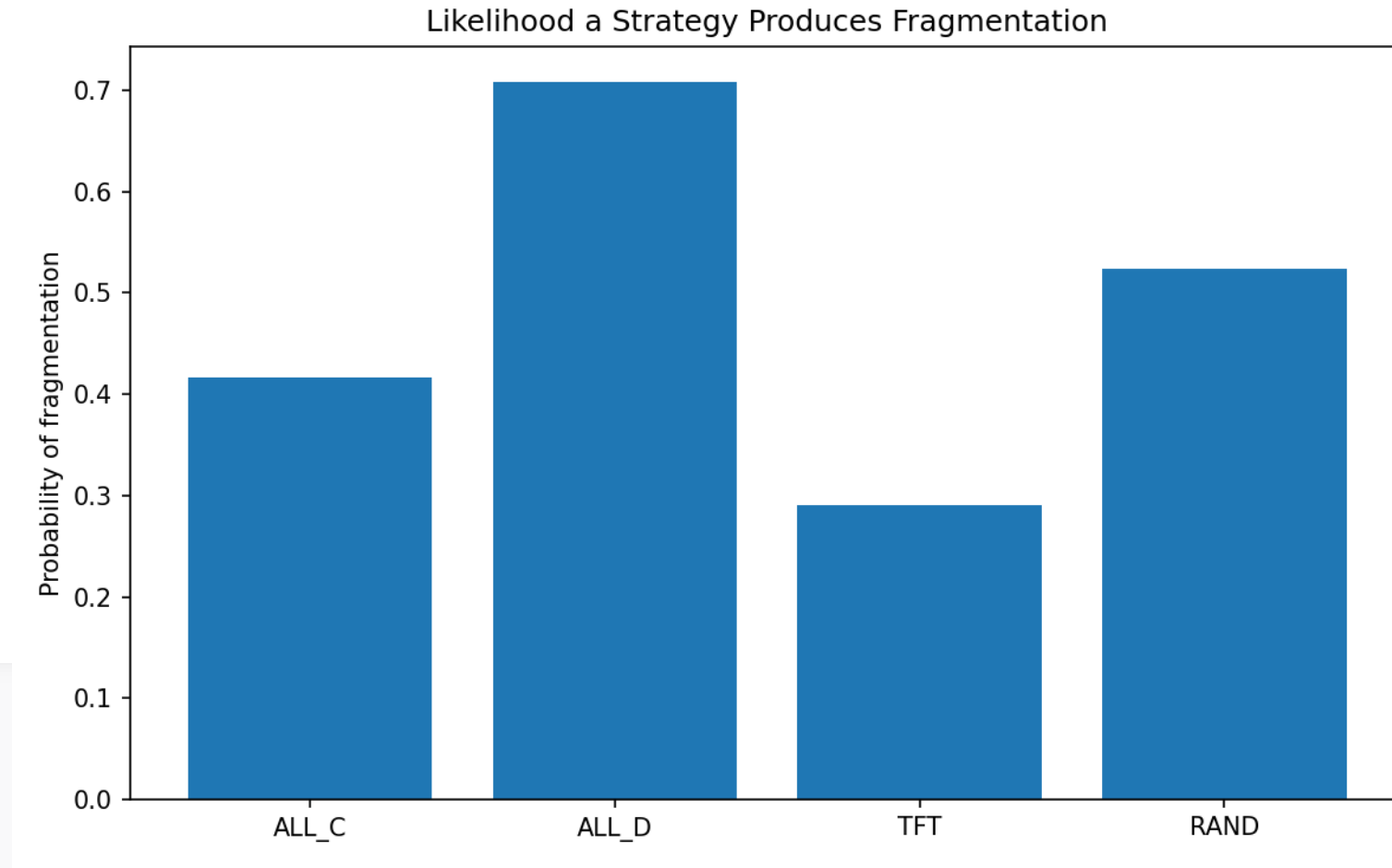
## 4.2 Emergence of Complex Behaviour: Fragmentation

- The group dynamics reveal a clear fragmentation pattern: units tend to break apart into smaller sub-groups as they expand and interact with their surroundings.
- Fragmentation is frequently examined as a mechanism that can undermine group cohesion and contribute to the eventual dissolution or extinction of armed groups.

([Hanne Fjelde](#) and [Desirée Nilsson](#) 2018) ([Kathleen Gallagher Cunningham](#), [Kristin M. Bakke](#), 2012)



## 4.2 Emergence of Complex Behaviour: Fragmentation

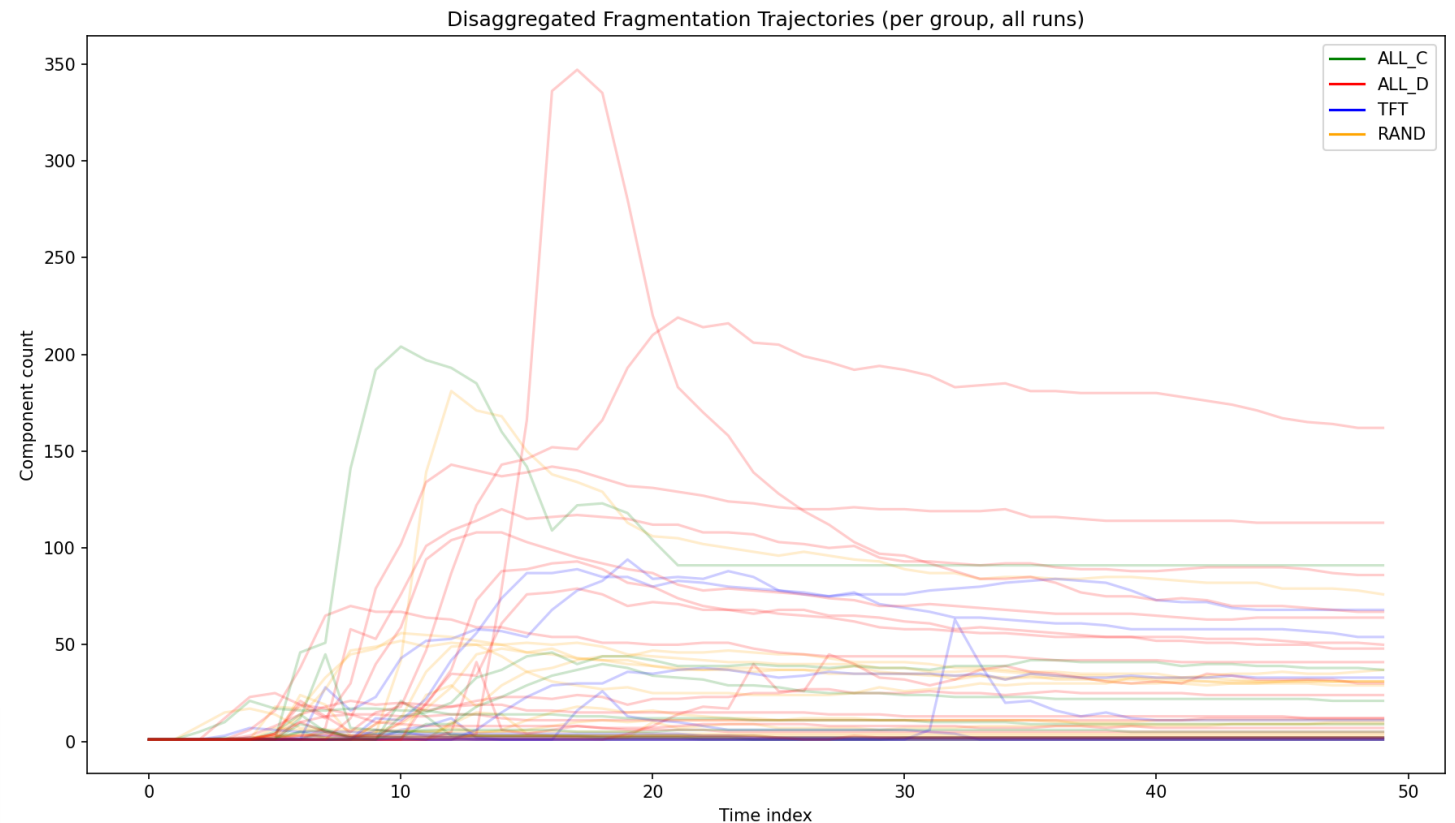




## 4.2 Emergence of Complex Behaviour: Fragmentation

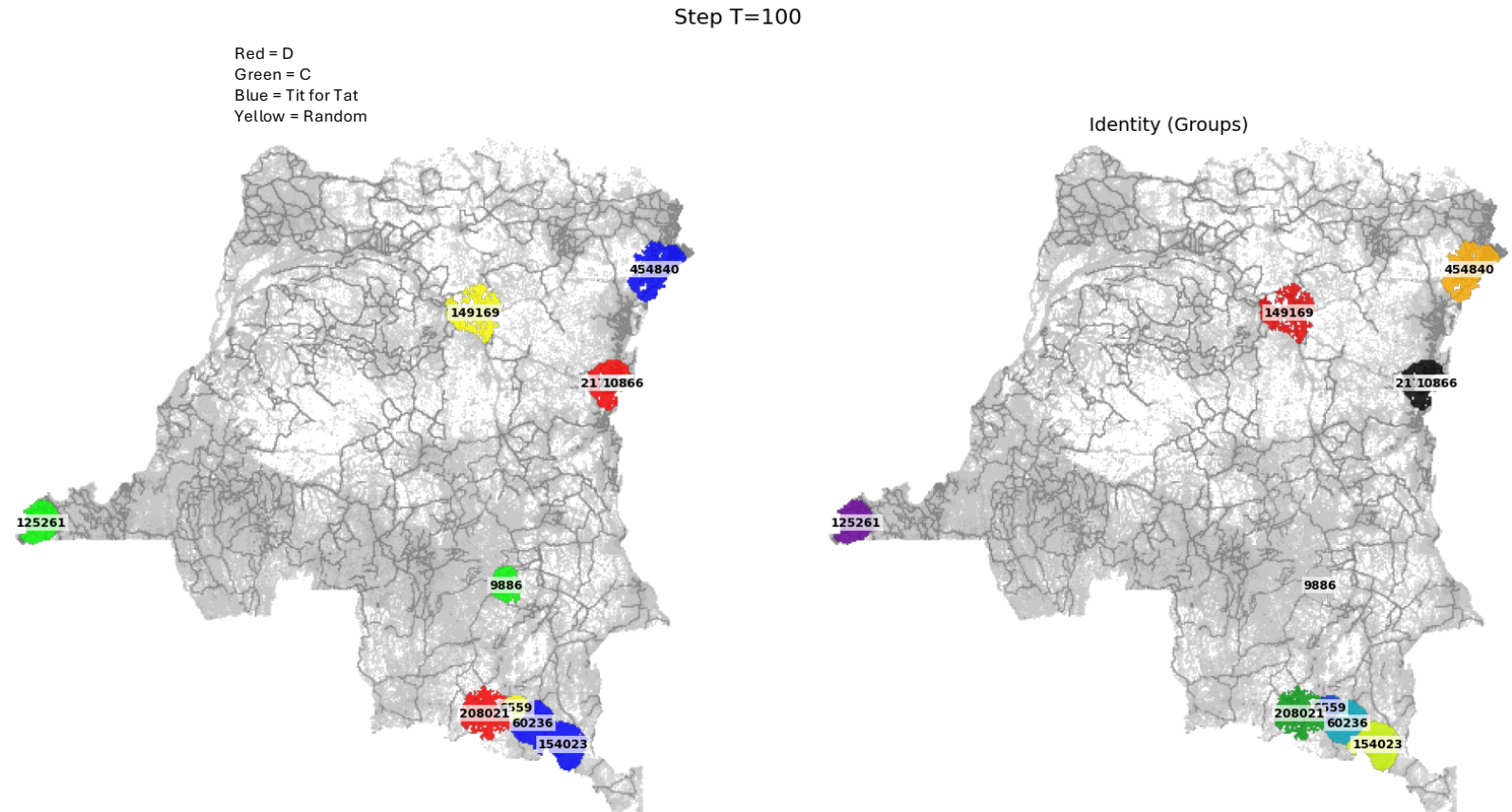
The plot shows for each initial group, how many groups stem from it, and how many remain at the end of the simulation.

→ At times, many small groups emerge, but most eventually die out.



## 4.2 Emergence of Complex Behaviour

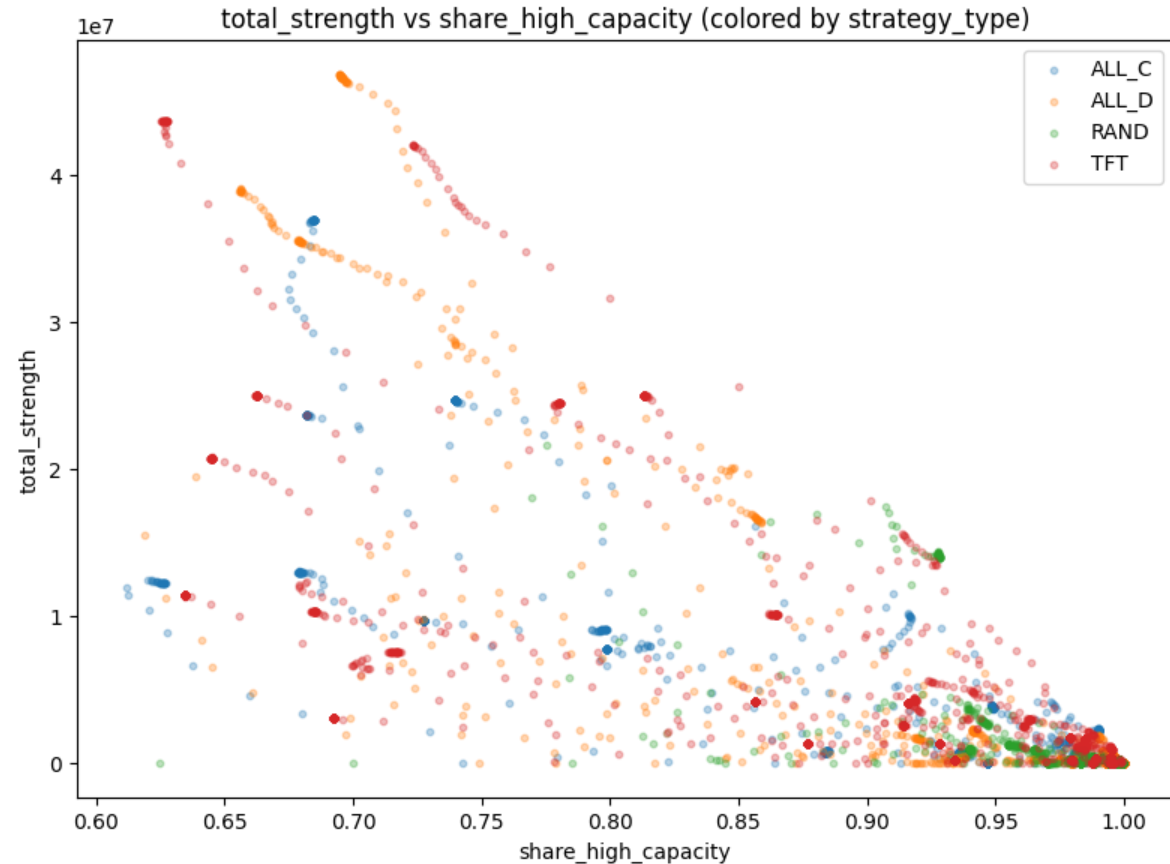
→ The model typically produces one dominant group that occupies the majority of the space, but resource heterogeneity enables several smaller groups to survive nonetheless.



## 4.2 Emergence of Complex Behaviour

- The **x-axis shows** the fraction of a group's mass located on high-resource terrain (*share\_high\_capacity*).
- The **y-axis shows** the group's total strength (*how large the group is*).
- Each point represents one group at one moment in time.
- Colors indicate the group's strategy type (*ALL\_C*, *ALL\_D*, *RAND*, *TFT*).

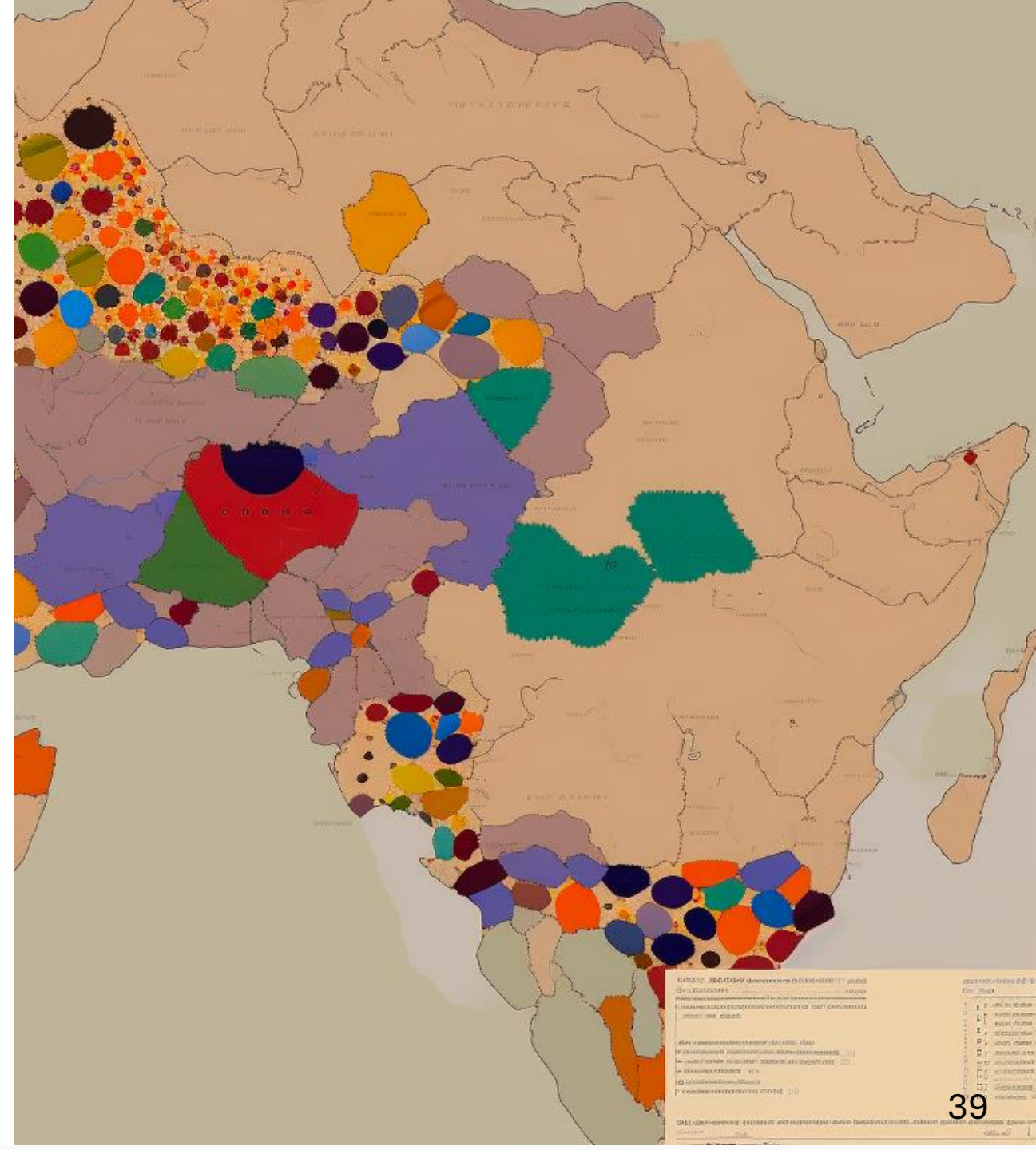
→ Interestingly, groups with extremely high *share\_high\_capacity* (0.90–1.00) are almost all weak.





# Limitations and future work

- Sensitivity analysis on Prisoner's Dilemma's values
- Sensitivity analysis over number of group



# Contributions

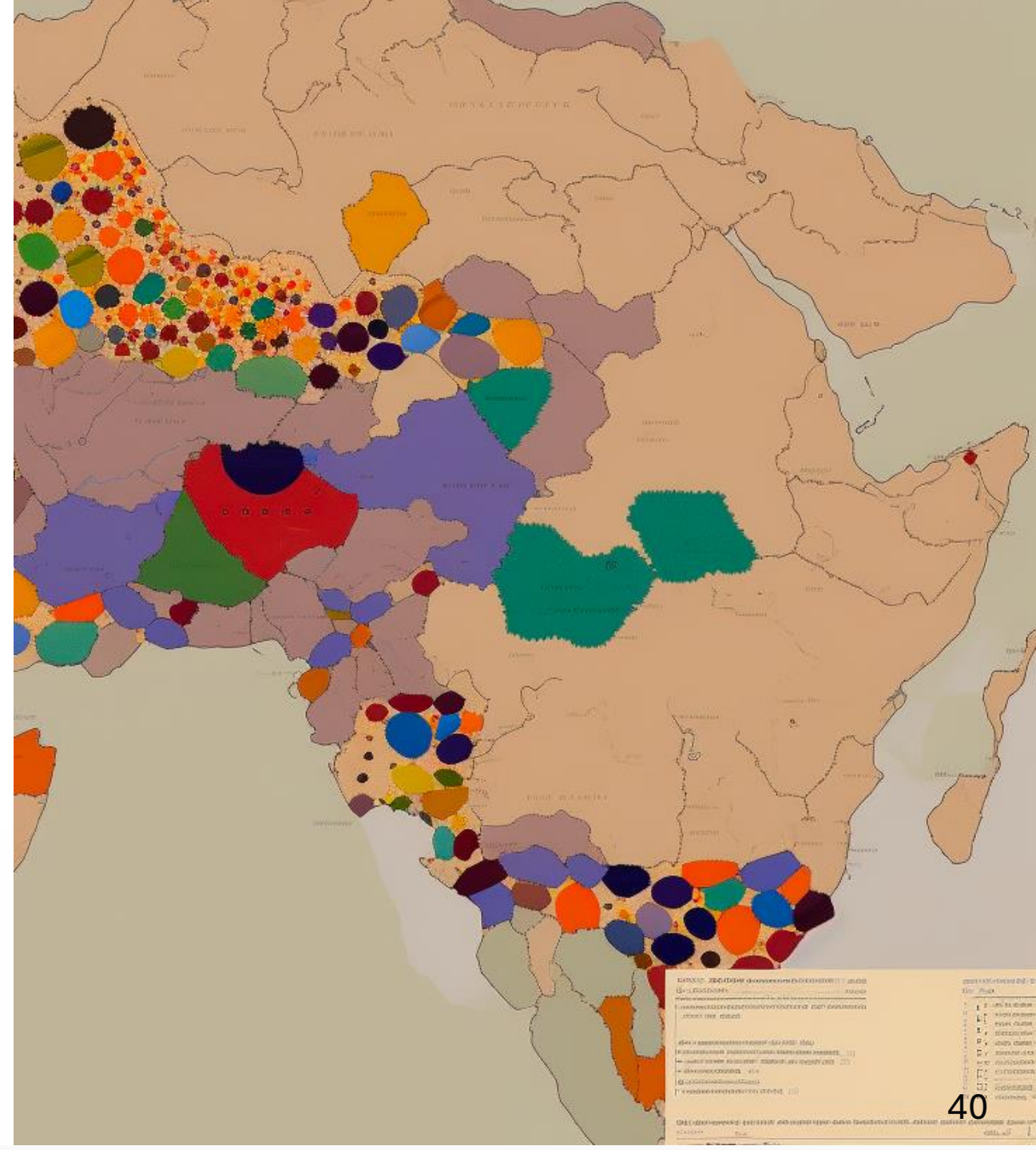
Francesco: Main Code

Lukas: Pure Resource Code

Vincent: Slides + Pure Resource Code Help

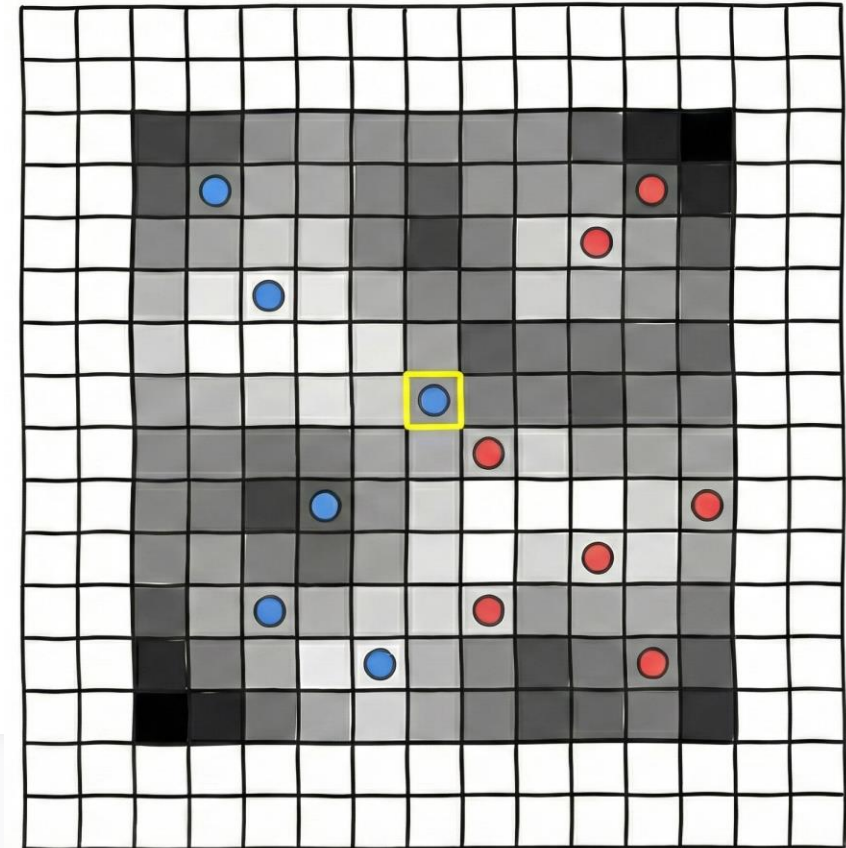
Vikram: Slides + Main Code Help

The research for the basis of model was evenly distributed



## A.1: Pure resource maximization

- Always **Defect (D)**
  - Always **Cooperate (C)**
  - Random (**R**)
  - Tit-for-Tat (**T**)
- 
- We also explore  
**pure resource maximization** strategies [\*]





## A.1: Pure resource maximization

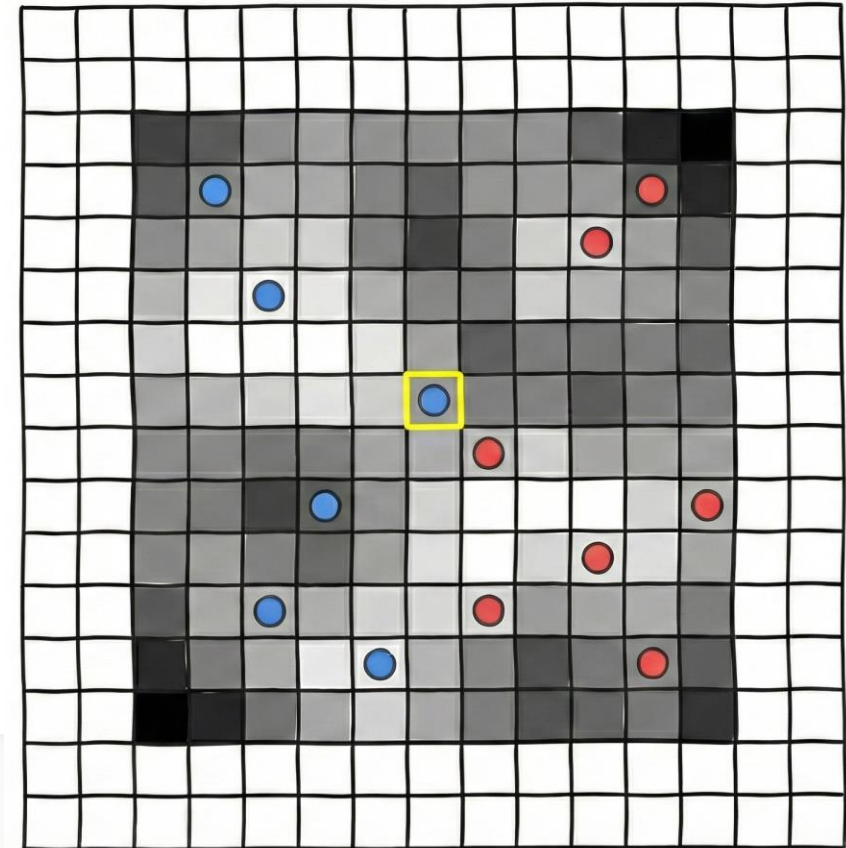
Dynamics are based on **pure resource maximization**

- **Interaction**

- Always **Defect (D)**
- Always **Cooperate (C)**
- Random (**R**)
- Tit-for-Tat (**T**)

- **Movement**

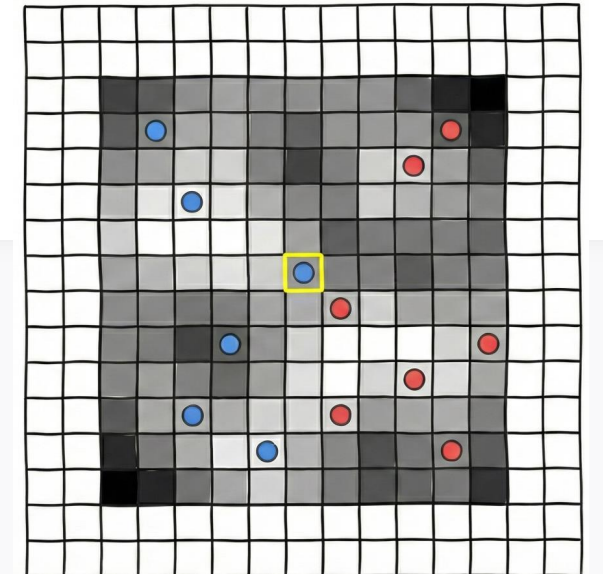
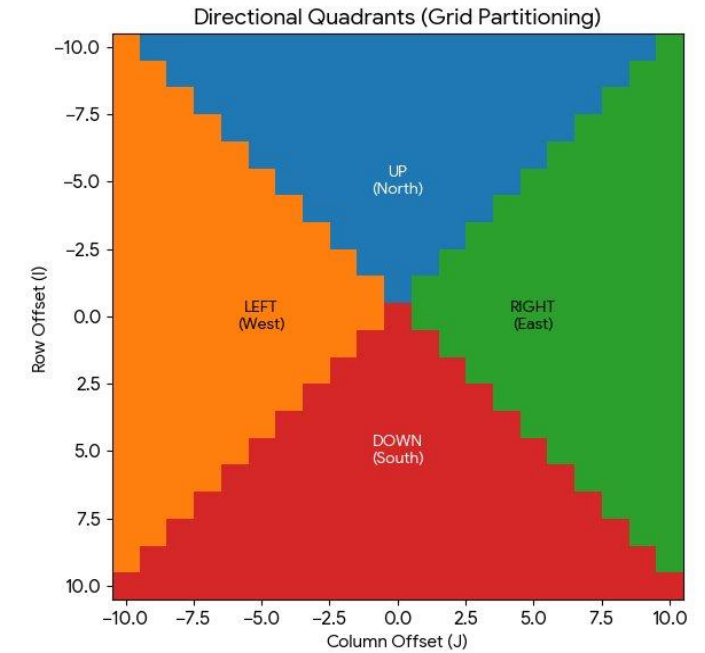
- Resource Attraction
- Opponent Detraction



## A.1: Movement: Unoccupied resources attract

- Unoccupied resources are always attracting

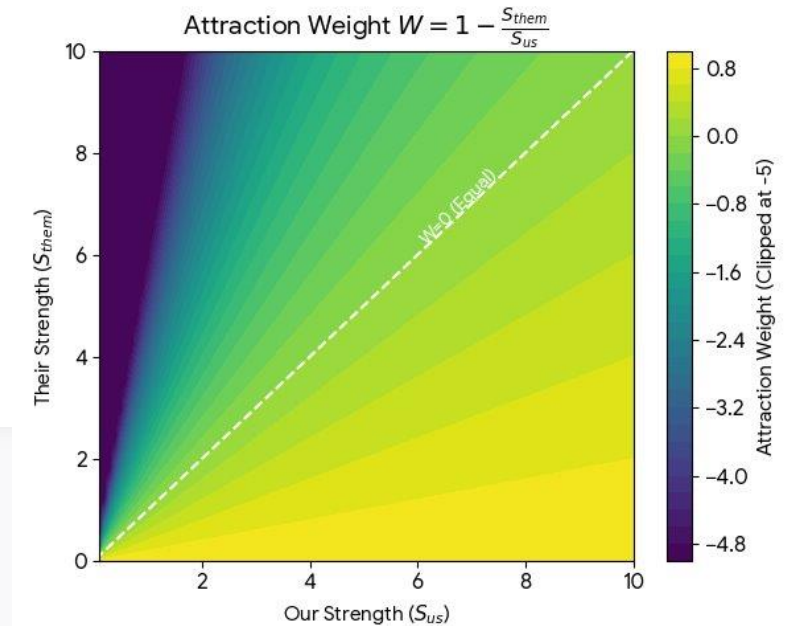
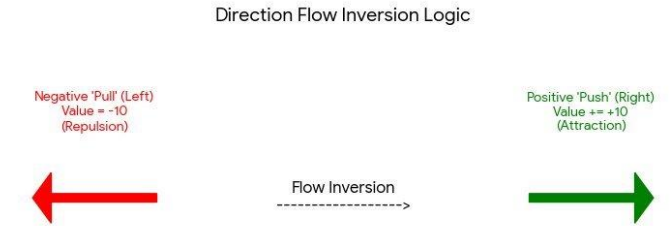
$$V_{dir} = \sum_{(y,x) \in Q_{dir}} K(y,x) \cdot \left[ R_{free} + \sum_{g \neq us} W(S_{us}, S_g) \cdot R_{occ}(g) \right]$$



# A.1: Movement: Opponents might detract

- Occupied resources attract/detract based on relative influence

$$V_{dir} = \sum_{(y,x) \in Q_{dir}} K(y,x) \cdot \left[ R_{free} + \sum_{g \neq u} \boxed{W(S_{us}, S_g)} \cdot R_{occ}(g) \right]$$





# A.1: Detraction by Strong Opponents: Movement Flow Inversion

- Negative action logits are inverted

$$Action_{Right} \leftarrow Action_{Right} - \min(0, Action_{Left})$$

Direction Flow Inversion Logic

Negative 'Pull' (Left)  
Value = -10  
(Repulsion)



Flow Inversion  
----->

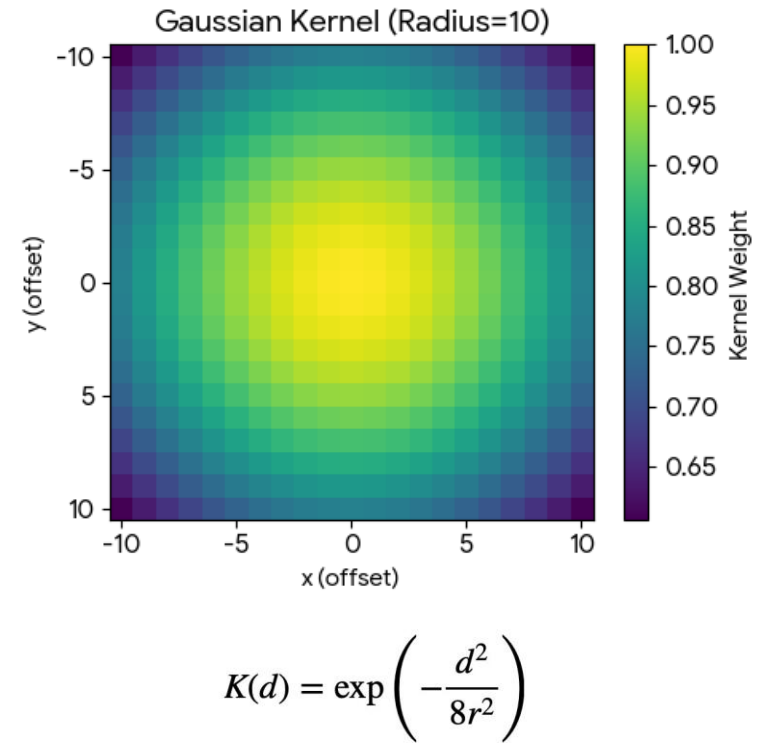
Positive 'Push' (Right)  
Value += +10  
(Attraction)



## A.1: Movement: Resource Depreciation by Distance

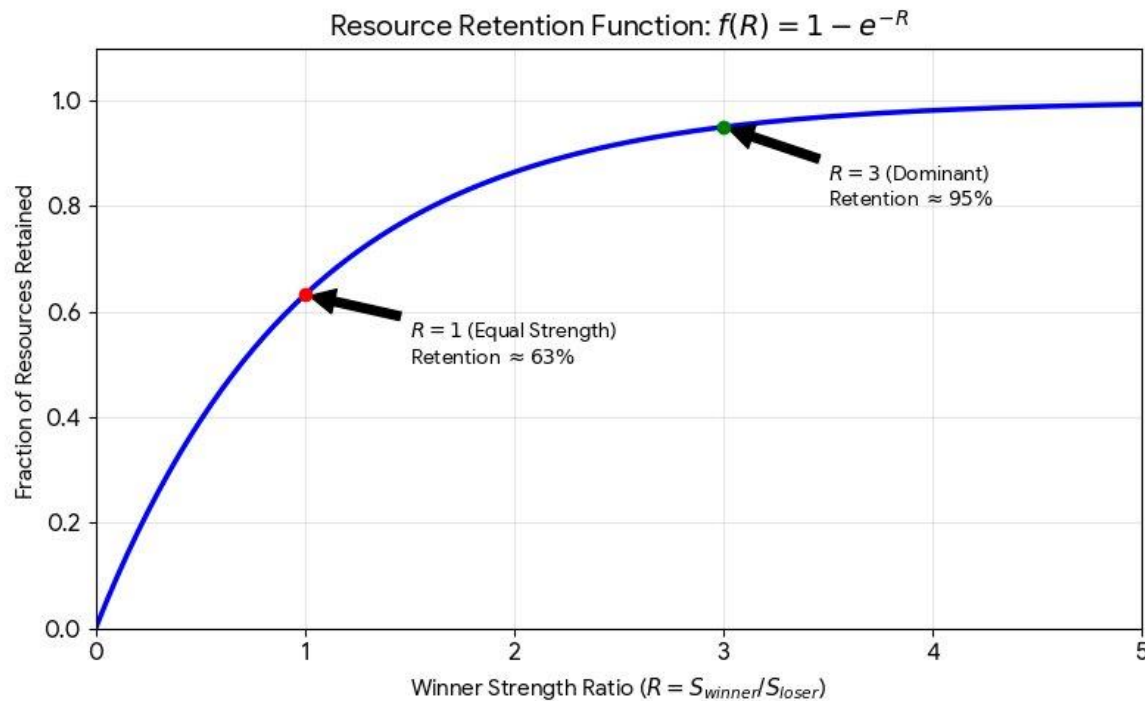
- Occupied resources are depreciated by distance

$$V_{dir} = \sum_{(y,x) \in Q_{dir}} K(y,x) \cdot \left[ R_{free} + \sum_{g \neq us} W(S_{us}, S_g) \cdot R_{occ}(g) \right]$$



# A.1: Resource Loss After Conflict

- Conflict decision is based on the **expected gain** after a conflict



$$P(\text{Win}_i) = \frac{S_i}{\sum_{j \in \text{Competitors}} S_j}$$

$$R = \frac{S_{\text{winner}}}{S_{\text{winner}} + S_{\text{expected\_opp}}}$$

$$O_{t+1} = (1 - \exp(-R)) \cdot O_t$$

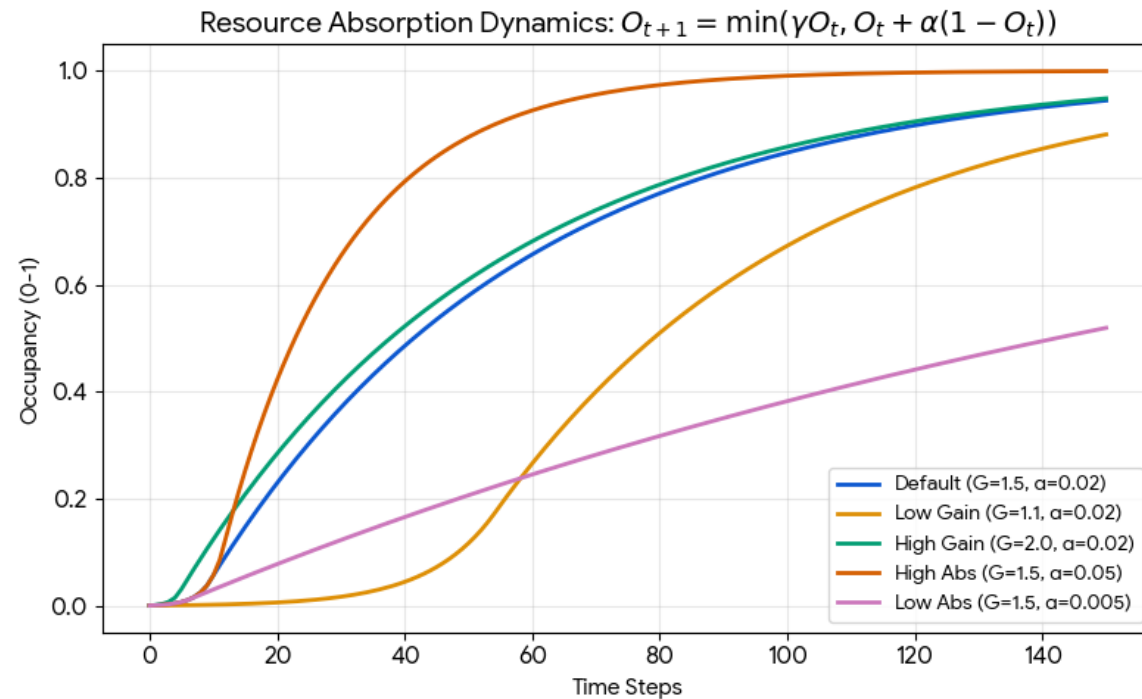
$$\bar{S}_{\text{other}} = \frac{\sum_{j \neq \text{us}} S_j}{N - 1}$$

$$S_{\text{opp\_expected}} = N_{\text{fights}} \cdot \bar{S}_{\text{other}}$$



## A.2: Resource Finiteness in Resource Absorption

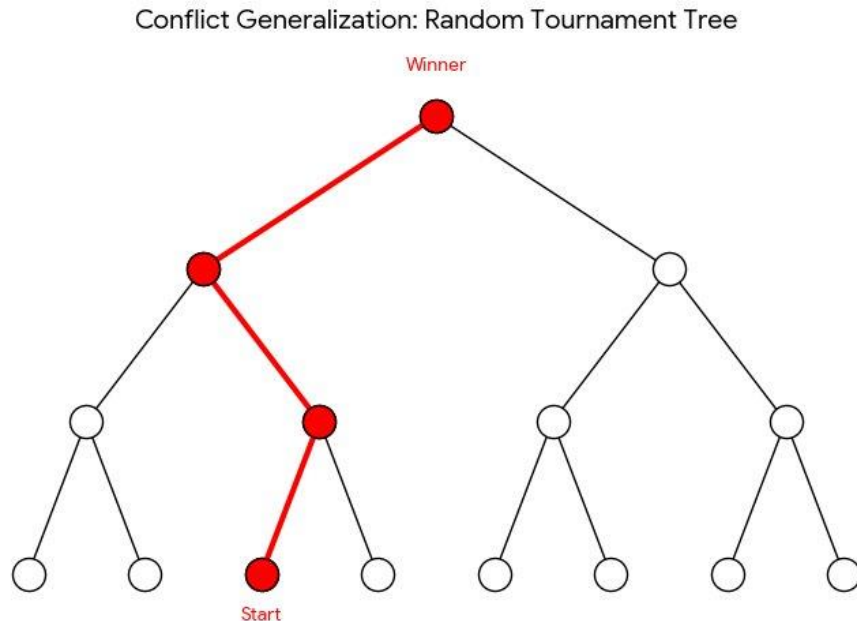
- We model **saturation** of resource absorption
- Groups are incentivized to continue expanding after saturation



$$O_{t+1} = \min \left( \gamma \cdot O_t, O_t + \alpha(1 - O_t) \right)$$

### A.3: Generalization to Multi-Group Conflicts

- For more than two groups, we model conflicts as a **random tournament**



$$E[\text{Gain}] = P(\text{win}) \cdot \Delta O_{\text{win}} + (1 - P(\text{win})) \cdot \Delta O_{\text{loss}}$$

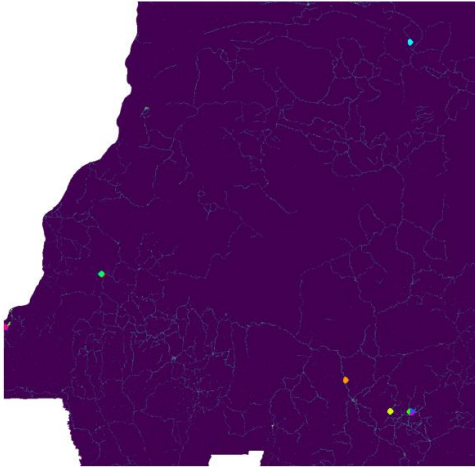
$$\text{Action} = \begin{cases} \text{Fight} & \text{if } E[\text{Gain}] > 0 \\ \text{Cooperate} & \text{if } E[\text{Gain}] \leq 0 \end{cases}$$

$$N_{\text{fights}} = \log_2(N_{\text{competitors}})$$

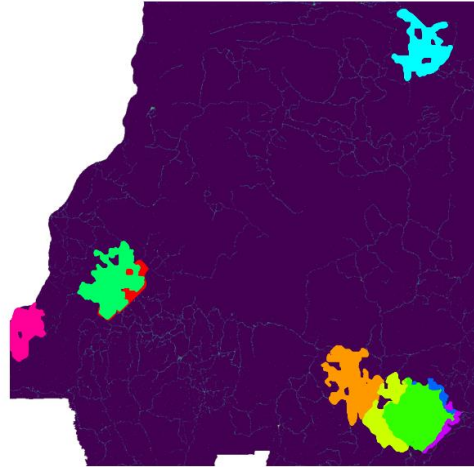
## A: Results: High absorption rate

- Groups spread according to unoccupied resources and opponent strength

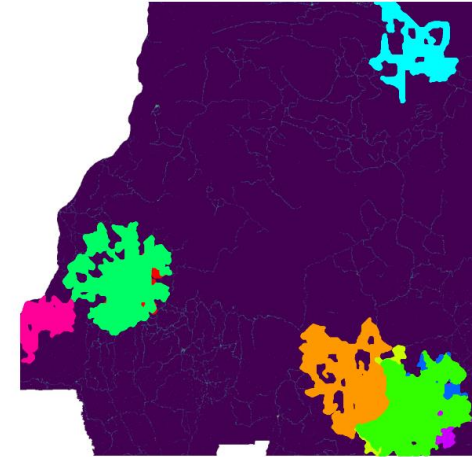
Resource/Group distribution, t=5



Resource/Group distribution, t=110



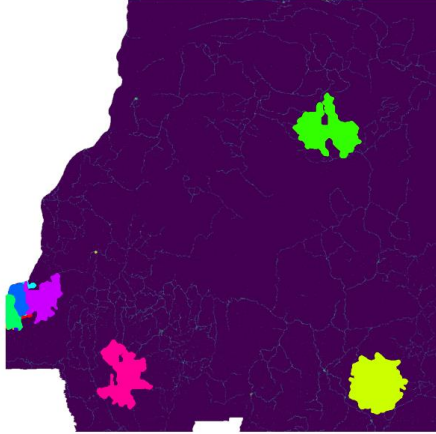
Resource/Group distribution, t=207



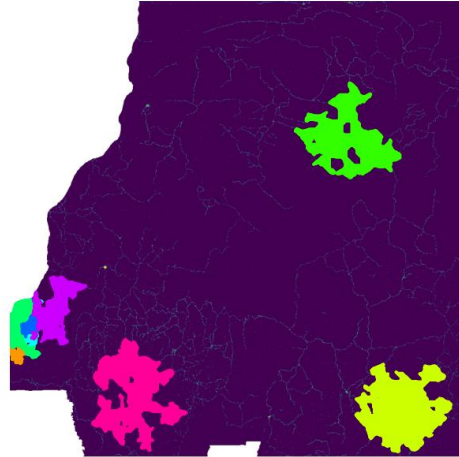


# A: Results: Low absorption

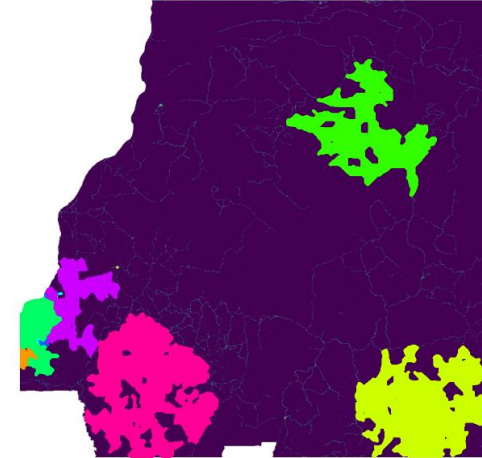
Resource/Group distribution,  $t=104$



Resource/Group distribution,  $t=194$



Resource/Group distribution,  $t=324$



# A: Glossary

| Variable        | Definition                                                        |
|-----------------|-------------------------------------------------------------------|
| $K(d)$          | Gaussian sensing kernel (decay over distance $d$ )                |
| $r$             | Sensing radius (neighborhood size)                                |
| $W$             | Attraction weight ( $> 0$ ) or repulsion ( $< 0$ ) score          |
| $S_{us}$        | Current strength of our group                                     |
| $S_{them}$      | Strength of a specific rival group                                |
| $V_{dir}$       | Aggregated utility value for moving in direction $dir$            |
| $Q_{dir}$       | Spatial quadrant mask (North, South, East, West)                  |
| $R_{free}$      | Unoccupied resources (always attractive)                          |
| $R_{occ}$       | Resources held by others (weighted by $W$ )                       |
| $O_t$           | Occupancy fraction of a cell at time $t$ (0.0 to 1.0)             |
| $\gamma$        | Max Occupancy Gain (limit on exponential growth)                  |
| $\alpha$        | Max Absorption Rate (limit on saturation speed)                   |
| $N$             | Number of competing groups in a single cell                       |
| $N_{fights}$    | Expected depth of conflict tournament ( $\approx \log_2 N$ )      |
| $S_{other}$     | Average strength of rival groups in the cell                      |
| $S_{exp}$       | Total expected opponent strength ( $N_{fights} \cdot S_{other}$ ) |
| $P(\text{Win})$ | Probability of winning conflict (based on strength)               |
| $R$             | Winner Strength Ratio (Winner vs. Losers)                         |